



Xger

Reference Manual

Version 2.53

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1 Overview

```
# Changelog
## 2.53
- **Changed**
  - `coderService` from 2.41 to 2.42
  - `coderConfig` from 2.43 to 2.44
  - `multiService` from 2.13 to 2.14
  - `lockStatus` from 1.3 to 1.4
  - `serviceStatus` from 2.42 to 2.43
  - `codedAudio` from 1.0 to 1.1
  - `rawAudio` from 1.1 to 1.2
  - `audioProfile` from 2.5 to 2.6
  - `audioStatus` from 2.14 to 2.15
  - `multiServiceProfile` from 2.5 to 2.6
  - `st2110Status` from 1.4 to 1.5
  - `sdiStatus` from 1.10 to 1.11

## 2.52
- **Changed**
  - `rawVideo` from 1.4 to 1.5
  - `videoProfile` from 2.10 to 2.11
  - `multiServiceProfile` from 2.4 to 2.5
  - `multiService` from 2.12 to 2.13
  - `coderService` from 2.40 to 2.41
  - `videoStatus` from 2.14 to 2.15
  - `serviceStatus` from 2.41 to 2.42
  - `expectedInput` from 1.4 to 1.5
  - `cardAllocation` from 1.7 to 1.8
  - `nmosStatusTypes` from 1.1 to 1.2
  - `cardStatus` from 1.8 to 1.9
  - `coderConfig` from 2.42 to 2.43
  - `vancProfile` from 1.6 to 1.7
  - `vancStatus` from 1.12 to 1.13

- **Added**
  - `dpi` version 1.0
  - `dpiStatus` version 1.0

## 2.51
- **Changed**
  - `videoProfile` from 2.9 to 2.10
  - `coderService` from 2.39 to 2.40
  - `multiService` from 2.11 to 2.12
  - `multiServiceProfile` from 2.3 to 2.4
  - `cardAllocation` from 1.6 to 1.7
  - `videoStatus` from 2.13 to 2.14
  - `expectedInput` from 1.3 to 1.4
  - `serviceStatus` from 2.40 to 2.41

## 2.50
- **Added**
  - `vpidInfo` version 1.0
  - `videoUsabilityInfo` version 1.1

- **Changed**
  - `coderService` from 2.38 to 2.39
  - `colorProfile` from 2.1 to 2.2
  - `colorComponent` from 1.0 to 1.1
  - `videoStatus` from 2.12 to 2.13
```

- `serviceStatus` from 2.39 to 2.40
- `sdiStatus` from 1.9 to 1.10
- `multiService` from 2.10 to 2.11
- `vancProfile` from 1.5 to 1.6
- `vancStatus` from 1.11 to 1.12
- `subtitling` from 1.1 to 1.2
- `tsSource` from 1.4 to 1.5

2.49

No changes

2.48

- ****Changed****
 - `audioProfile` from 2.4 to 2.5
 - `audioStatus` from 2.13 to 2.14
 - `coderConfig` from 2.39 to 2.40
 - `coderService` from 2.37 to 2.38
 - `multiService` from 2.9 to 2.10
 - `multiServiceProfile` from 2.2 to 2.3
 - `serviceStatus` from 2.38 to 2.39
 - `st2110Status` from 1.3 to 1.4
 - `vancStatus` from 1.10 to 1.11

2.47

- ****Added****
 - 'scte35LogApi' version 1.0
 - sdiRouterConfig version 1.1
- ****Changed****
 - `vancStatus` from 1.9 to 1.10
 - `serviceStatus` from 2.37 to 2.38

2.46

- ****Added****
 - `nmosStatusTypes` version 1.1
- ****Changed****
 - `nmosConfig` from 1.0 to 1.1
 - `cardConfig` from 1.2 to 1.3
 - `videoProfile` from 2.8 to 2.9
 - `expectedInput` from 1.2 to 1.3
 - `multiServiceProfile` from 2.1 to 2.2
 - `cardAllocation` from 1.5 to 1.6
 - `serviceStatus` from 2.36 to 2.37
 - `videoStatus` from 2.11 to 2.12
 - `coderService` from 2.36 to 2.37
 - `multiService` from 2.8 to 2.9
 - `cardStatus` from 1.7 to 1.8

2.45

- `multiService` from 2.7 to 2.8
- `coderService` from 2.35 to 2.36
- `tsSource` from 1.3 to 1.4
- `serviceStatus` from 2.35 to 2.36
- `vancStatus` from 1.8 to 1.9

2.44

- ****Changed****
 - `multiService` from 2.6 to 2.7
 - `coderService` from 2.34 to 2.35
 - `tsSource` from 1.2 to 1.3
 - `ipStream` from 1.2 to 1.3
 - `st2110Types` from 1.0 to 1.2

```
- `serviceStatus` from 2.34 to 2.35
- `st2110Status` from 1.2 to 1.3
- `ipConnection` from 1.4 to 1.5
- `ipService` from 1.1 to 1.2
- `rawAudio` from 1.0 to 1.1
- `audioProfile` from 2.3 to 2.4
- `multiServiceProfile` from 2.0 to 2.1
- `audioStatus` from 2.12 to 2.13
- `vancStatus` from 1.7 to 1.8
- `vancProfile` from 1.4 to 1.5

## 2.43
- **Added**
  - `logoInsertion` 1.0
- **Changed**
  - `coderService` from 2.33 to 2.34
  - `multiService` from 2.5 to 2.6
  - `audioStatus` from 2.11 to 2.12
  - `sdiStatus` from 1.8 to 1.9
  - `serviceStatus` from 2.33 to 2.34

## 2.42
- **Changed**
  - `serviceStatus` from 2.32 to 2.33
  - `vancStatus` from 1.6 to 1.7
  - `cardConfig` from 1.1 to 1.2
- **Added**
  - `nmosConfig` 1.0
  - `nmosLabelConfig` 1.0

## 2.41
- **Changed**
  - `vancProfile` from 1.3 to 1.4
  - `coderService` from 2.32 to 2.33
  - `vancStatus` from 1.5 to 1.6
  - `multiService` from 2.4 to 2.5
  - `serviceStatus` from 2.31 to 2.32

## 2.40
- **Changed**
  - `coderService` from 2.31 to 2.32
  - `multiService` from 2.3 to 2.4
  - `ipInterface` from 1.4 to 1.5
  - `slotIpInterface` from 1.4 to 1.5
  - `linkModes` from 1.1 to 1.2

## 2.39
- **Added**
  - `poisServerStatus` version 1.0

- **Changed**
  - `poolCtrlAllocs` from 1.0 to 1.1
  - `lockStatus` from 1.2 to 1.3
  - `coderService` from 2.30 to 2.31
  - `serviceStatus` from 2.30 to 2.31
  - `multiService` from 2.2 to 2.3

## 2.38
- **Added**
  - `esamConfig` version 1.0
```

```
- **Changed**
- `multiService` from 2.1 to 2.2
- `poolConfig` from 1.1 to 1.2
- `coderService` from 2.29 to 2.30

## 2.37
- **Changed**
- `coderService` from 2.28 to 2.29
- `multiService` from 2.0 to 2.1
- `dejitterTypes` from 1.0 to 1.1
- `ipConnection` from 1.3 to 1.4
- `ipService` from 1.0 to 1.1
- `videoStatus` from 2.10 to 2.11
- `serviceStatus` from 2.29 to 2.30
- `psiTypes` from 1.5 to 1.6
- `psiStatus` from 1.5 to 1.6
- `inputRedundancyStatus` from 1.2 to 1.3

## 2.36
- **Changed**
- `coderService` from 2.27 to 2.28
- `multiServiceProfile` from 1.10 to 2.0
- `multiService` from 1.22 to 2.0
- `ipConnection` from 1.2 to 1.3
- `serviceStatus` from 2.28 to 2.29
- `sdpUpload` from 1.0 to 1.1
- **Added**
- `ipStreamUtils` version 0.0
- `ipService` version 1.0

## 2.35
- **Changed**
- `coderService` from 2.26 to 2.27
- `multiService` from 1.21 to 1.22
- `sdiStatus` from 1.7 to 1.8
- `serviceStatus` from 2.27 to 2.28

## 2.34
- **Changed**
- `redundancyGroupStatus` from 1.1 to 1.2;

## 2.33
- **Added**
- `slateInsertion` 1.0
- `imageUpload` 1.0
- **Changed**
- `multiService` from 1.20 to 1.21
- `coderService` from 2.25 to 2.26
```

2 afd (1.0)

2.1 Type Reference

2.1.1 AfdCode

enum

UNKNOWN

RESERVED_1

BOX_16_9_TOP

BOX_14_9_TOP

BOX_16_9_CENTRE

RESERVED_5

RESERVED_6

RESERVED_7

SAME

_4_3_CENTRE

_16_9_CENTRE

_14_9_CENTRE

RESERVED_12

_4_3_PROTECT_14_9_CENTRE

_16_9_PROTECT_14_9_CENTRE

_16_9_PROTECT_4_3_CENTRE

3 alternateInputStatus (1.4)

3.1 Type Reference

3.1.1 AlternateInputStatus

- When alternative input is used, this struct holds the lists of IpStreamStatus of both input paths.
- It also holds the status of the alternative or seamless input configuration.
-

struct

path1	list of IpStreamStatus.IpStreamStatus Status of the IpStreams on path 1. •
path2	list of IpStreamStatus.IpStreamStatus Status of the IpStreams on path 2. •
seamlessType	list of SeamlessTypeStatus list of statuses of the detected type of seamless input.

3.1.2 NearSeamlessStatus

Status of a near seamless input.

struct

active	Path The currently active path. Only used for near seamless.
synchronized	bool True if the inputs are synchronized.

3.1.3 Path

The path of a near seamless input which a status describes.

enum

path1	
path2	

3.1.4 SeamlessTypeStatus

Status of a SMPTE 2022-7 Seamless input or a near seamless input.

variant

smp2022_7	Smp2022_7Status Status of a SMPTE 2022-7 Seamless input.
	NearSeamlessStatus

`nearSeamless`

Status of a near seamless input.

3.1.5 Smppte2022_7Status

Status of a SMPTE 2022-7 Seamless input.

struct

<code>id</code>	UUID The id of the IpStream this status represents.
<code>synchronized</code>	bool True if the inputs are synchronized.
<code>relativeDelay</code>	int Delay between path A and path B in milliseconds.
<code>margin</code>	int Difference between the delay buffer size and the current relative delay in milliseconds. It is what remains of the buffer.
<code>maxRelativeDelay</code>	int The size of the delay buffer in milliseconds.

4 aspectRatio (1.0)

4.1 Type Reference

4.1.1 AspectRatioStandard

enum

SMPTE_2016

WSS

VIDEO_INDEX

5 audioProfile (2.6)

5.1 Overview

```
# Changelog
## 2.6
- **Changed**
  - `codedAudio` from 1.0 to 1.1
  - `rawAudio` from 1.1 to 1.2

## 2.5
- **Added**
  - followInputChannelMode as optional(bool) in `AudioProfileStruct`

## 2.4
- **Changed**
  - `rawAudio` from 1.0 to 1.1
```

5.2 Command Reference

5.2.1 GetAudioProfiles

- message **GetAudioProfiles.Request**
- message **GetAudioProfiles.Response**
- message **GetAudioProfiles.Failure**

5.2.2 SetAudioProfiles

- message **SetAudioProfiles.Request**
- message **SetAudioProfiles.Response**
- message **SetAudioProfiles.Failure**

5.2.3 DeleteAudioProfiles

- message **DeleteAudioProfiles.Request**
- message **DeleteAudioProfiles.Response**
- message **DeleteAudioProfiles.Failure**

5.3 Type Reference

5.3.1 AacContainer

enum

ADTS

LATM

5.3.2 AacLcParameters

struct

container

AacContainer

5.3.3 AudioProfile

AudioProfileStruct

5.3.4 AudioProfileStruct

struct

label	string
codec	codedAudio.AudioCodec
channelMode	rawAudio.ChannelMode
followInputChannelMode	optional bool
sampleRate	SampleRate
bitRate	int
cparams	CodecParameters

5.3.5 CodecParameters

variant

mpeg1l2	Mpeg1l2Parameters
aac	AacLcParameters
dd	DolbyDigitalPlusParameters

5.3.6 CompProfileAndDialNorm

struct

dialnorm	int
compProfile	DdppCompressionProfile

5.3.7 DdppAdConverterType

enum

Undefined
Standard
HDCD

5.3.8 DdppBitstreamMode

enum

Undefined
CompleteMain
MusicAndEffects
VisuallyImpaired
HearingImpaired
Dialogue
Commentary

Emergency
VoiceOver
Karaoke

5.3.9 DdppChannelModeParameters

variant

mono	DdppChannelModeParameters.mono
stereo	DdppStereoParameters
multi	DdppMultiChannelParameters

5.3.10 DdppChannelModeParameters.mono

empty **struct**

5.3.11 DdppCompressionProfile

struct

dynrng	DdppDrcProfile
compr	DdppDrcProfile

5.3.12 DdppDownmixLevel

enum

Undefined
Pos_3Db
Pos_1_5Db
Pos_0Db
Neg_1_5Db
Neg_3Db
Neg_4_5Db
Neg_6Db
NegInfDb

5.3.13 DdppDownmixMode

enum

Undefined
NotIndicated
LtRt
LoRo
ProLogicII

5.3.14 DdppDrcProfile

enum

Undefined

None

FilmStandard

FilmLight

MusicStandard

MusicLight

Speech

5.3.15 DdppEncExtMd

struct

extMetadataProgramNum	int
extMetadataRevMode	DdppExtMetadataReversionMode
extMetadataOverwriteMode	DdppExtMetadataOverwriteMode

5.3.16 DdppEncManualMd

struct

dialnorm	int
bsmod	DdppBitstreamMode
compProfile	DdppCompressionProfile
mixlevel	int
roomtyp	DdppRoomType
copyrightb	bool
origbs	bool
adconvtyp	DdppAdConverterType
evolutionFrameworkEnabled	bool
cmParams	DdppChannelModeParameters

5.3.17 DdppExtMetadataOverwriteMode

variant

none	DdppExtMetadataOverwriteMode.none
dialnorm	int
compProfile	DdppCompressionProfile
compProfileAndDialNorm	CompProfileAndDialNorm

5.3.18 DdppExtMetadataOverwriteMode.none

empty struct

5.3.19 DdppExtMetadataOverwriteModeEnum

enum

NONE

DIALNORM

COMPPROFILE

COMPPROFILEANDDIALNORM

5.3.20 DdppExtMetadataReversionMode

enum

Undefined

LastUsedMetadata

LastHostSettings

5.3.21 DdppHeadphoneMode

enum

Undefined

NotIndicated

Disabled

Enabled

5.3.22 DdppMetadata

variant

encManualMd

DdppEncManualMd

encExtMd

DdppEncExtMd

transMd

DdppTransMd

5.3.23 DdppMetadataEnum

enum

DDPPENCMANUALMD

DDPPENCEXTMD

DDPPTRANSMMD

5.3.24 DdppMultiChannelParameters

struct

lfelpfon

bool

sur90on

bool

suratton

bool

lorocmixlev

DdppDownmixLevel

lorosurmixlev	DdppDownmixLevel
ltrtcmixlev	DdppDownmixLevel
ltrtsurmixlev	DdppDownmixLevel
dmixmod	DdppDownmixMode
dsurexmod	DdppSurroundMode

5.3.25 DdppRoomType

enum

Undefined
NotIndicated
LargeRoom
SmallRoom

5.3.26 DdppStereoParameters

struct

dsurmod	DdppSurroundMode
---------	------------------

5.3.27 DdppSurroundMode

enum

Undefined
NotIndicated
Disabled
Enabled

5.3.28 DdppTransMd

struct

dialnorm	int
evolutionFrameworkEnabled	bool

5.3.29 DeleteAudioProfiles.Failure

empty **struct**

5.3.30 DeleteAudioProfiles.Request

struct

ids	list of UUID
-----	--------------

5.3.31 DeleteAudioProfiles.Responseempty **struct****5.3.32 DolbyDigitalPlusParameters****struct**metadata **DdppMetadata****5.3.33 GetAudioProfiles.Failure**empty **struct****5.3.34 GetAudioProfiles.Request**empty **struct****5.3.35 GetAudioProfiles.Response****struct**

data	map from UUID to AudioProfile
info	GetAudioProfiles.Response.info

5.3.36 GetAudioProfiles.Response.info**struct**totalEntries **int****5.3.37 Mpeg1l2Parameters**empty **struct****5.3.38 SampleBitDepth****enum**

SBD_16
SBD_20
SBD_24

5.3.39 SampleRate**enum**

SR_48kHz

5.3.40 SetAudioProfiles.Failure

empty **struct**

5.3.41 SetAudioProfiles.Request

struct

data **map** from **UUID** to **AudioProfile**

5.3.42 SetAudioProfiles.Response

empty **struct**

6 audioStatus (2.15)

6.1 Overview

```
# Changelog
## 2.15
- **Changed**
  - `audioProfile` from 2.5 to 2.6
  - `codedAudio` from 1.0 to 1.1
  - `rawAudio` from 1.1 to 1.2

## 2.14
- **Changed**
  - `audioProfile` from 2.4 to 2.5

## 2.13
- **Changed**
  - `rawAudio` from 1.0 to 1.1
  - `audioProfile` from 2.3 to 2.4

## 2.12
- **Changed**
  - `embeddedChannel` to `embeddedChannels` in `BasicConfig`; now a list of all channels in use a

## 2.11
- **Changed**
  - `component` from 1.3 to 1.5
```

6.2 Type Reference

6.2.1 Audio

struct

channelId	int
uid	int
source	Source
output	Output
incorrectDspSpeed	optional bool

6.2.2 AudioSet

struct

main	Audio
backup	optional Audio
active	Input

6.2.3 AudioStatus

struct

audios	list of AudioSet
--------	------------------

6.2.4 BasicConfig

struct

codec	optional codedAudio.AudioCodec
componentType	optional component.ComponentType
sampleRate	optional audioProfile.SampleRate
bitRate	optional int
channelMode	optional rawAudio.ChannelMode
numChannels	optional int
aacContainer	optional audioProfile.AacContainer
embeddedChannels	optional list of int

6.2.5 CondensedAudio

struct

channelId	int
uid	int
source	CondensedAudio.source
output	CondensedAudio.output

6.2.6 CondensedAudio.output

struct

codec	optional codedAudio.AudioCodec
channelMode	optional rawAudio.ChannelMode
bitRate	optional int
ddpp	optional CondensedAudio.output.ddpp

6.2.7 CondensedAudio.source

struct

addBsiByte0	optional int
-------------	--------------

6.2.8 CondensedAudioStatus

struct

audios	list of CondensedAudio
--------	-------------------------------

6.2.9 DdppAcMod

enum

_1_0
_1_1
_2_0

_2_1	
_2_2	
_3_0	
_3_1	
_3_2	
_3_3	
_3_4	

6.2.10 DdppDec

struct

outputchannelconfig	rawAudio.ChannelMode
drcmode	DdppDecDrcMode
drchighcut	DdppDecDrcScale
drclowboost	DdppDecDrcScale
stereodownmixmode	DdppDecStereoDownmixMode

6.2.11 DdppDecDrcMode

enum

Undefined	
Custom_0	
Line	
Rf	
portable_Neg_8_Db	
portable_Neg_11_Db	
portable_Neg_14_Db	

6.2.12 DdppDecDrcScale

enum

Undefined	
Min	
Max	

6.2.13 DdppDecStereoDownmixMode

enum

Undefined	
Auto	
LtRt	
LoRo	

6.2.14 DdppEnc

struct

dialnorm	int
bsmod	audioProfile.DdppBitstreamMode
dynrng	audioProfile.DdppDrcProfile
compr	audioProfile.DdppDrcProfile
dsurmod	audioProfile.DdppSurroundMode
mixlevel	int
roomtyp	audioProfile.DdppRoomType
copyrightb	YesNo
origbs	YesNo
lorocmixlev	audioProfile.DdppDownmixLevel
lorosurmixlev	audioProfile.DdppDownmixLevel
ltrtcmixlev	audioProfile.DdppDownmixLevel
ltrtsurmixlev	audioProfile.DdppDownmixLevel
dmixmod	audioProfile.DdppDownmixMode
dsurexmod	audioProfile.DdppSurroundMode
dheadphonmod	audioProfile.DdppHeadphoneMode
adconvtyp	audioProfile.DdppAdConverterType
lfelpfon	OnOff
sur90on	OnOff
suratton	OnOff
evolutionFrameworkEnabled	OnOff

6.2.15 DdppMetadata

struct

acmod	DdppAcMod
lfeon	OnOff
datarate	optional int
samplerate	optional int
dialnorm	int
dialnorm2	optional int
bsmod	audioProfile.DdppBitstreamMode
dsurmod	audioProfile.DdppSurroundMode
mixlevel	optional int
roomtyp	optional audioProfile.DdppRoomType
copyrightb	YesNo
origbs	YesNo
cmixlev	optional audioProfile.DdppDownmixLevel
surmixlev	optional audioProfile.DdppDownmixLevel
lorocmixlev	audioProfile.DdppDownmixLevel
lorosurmixlev	audioProfile.DdppDownmixLevel
ltrtcmixlev	audioProfile.DdppDownmixLevel
ltrtsurmixlev	audioProfile.DdppDownmixLevel
dmixmod	audioProfile.DdppDownmixMode
dsurexmod	audioProfile.DdppSurroundMode

dheadphonmod	optional audioProfile.DdppHeadphoneMode
adconvtyp	audioProfile.DdppAdConverterType
adconvtyp2	optional audioProfile.DdppAdConverterType
addbsie	optional YesNo
addbsibyte0	optional int

6.2.16 DepMetadata

struct

frameRate	float
numberOfPrograms	int
programConfig	DolbyEProgramConfig
bitDepth	int
programSelect	optional int

6.2.17 DolbyDigitalPlusMetEvoStatus

enum

Nofound
 ErrorParse
 ErrorPcmhash
 ErrorSerialize
 valid

6.2.18 DolbyEProgramConfig

enum

_51_2
 _51_1_1
 _4_4
 _4_2_2
 _4_2_1_1
 _4_1_1_1_1
 _2_2_2_2
 _2_2_2_1_1
 _2_2_1_1_1_1
 _2_1_1_1_1_1_1
 _1_1_1_1_1_1_1_1
 _51
 _4_2
 _4_1_1
 _2_2_2
 _2_2_1_1
 _2_1_1_1_1
 _1_1_1_1_1_1_1
 _4

_2_2	
_2_1_1	
_1_1_1_1	
_71	
_71_Screen	

6.2.19 EvolutionFrameworkMetadata

struct

evoPresent	YesNo
evoStatus	DolbyDigitalPlusMetEvoStatus

6.2.20 Input

enum

MAIN	
BACKUP	

6.2.21 LongTermLoudness

struct

loudnessLevel	optional float
maxTruePeak	optional float
gain	float

6.2.22 Loudness

struct

ltll	LongTermLoudness
stll	optional YesNo
peakLimiter	optional YesNo

6.2.23 OnOff

enum

Undefined	
On	
Off	

6.2.24 Output

struct

pid	optional int
-----	--------------

basicConfig	BasicConfig
advancedConfig	optional DdppEnc
cannotEncode	bool
loudness	optional Loudness

6.2.25 Process

enum

ENCODE	
TRANSCODE	
DECODE	
PASSTHROUGH	

6.2.26 Source

struct

process	Process
pid	optional int
basicConfig	BasicConfig
advancedConfig	optional DdppDec
bitstreamMetadata	Source.bitstreamMetadata
cannotDecode	optional bool

6.2.27 Source.bitstreamMetadata

struct

dep	optional DepMetadata
ddpp	optional DdppMetadata
evo	optional EvolutionFrameworkMetadata

6.2.28 YesNo

enum

Undefined	
Yes	
No	

7 cardAllocation (1.8)

7.1 Overview

Changelog:

1.8

- Changed
 - expectedInput from 1.4 to 1.5

1.7

- Changed
 - expectedInput from 1.3 to 1.4

1.6

- Changed
 - expectedInput from 1.2 to 1.3

1.5

- Changed
 - expectedInput from 1.1 to 1.2

7.2 Command Reference

7.2.1 GetCardAllocations

- message **GetCardAllocations.Request**
- message **GetCardAllocations.Response**

7.2.2 AssessAllocationPossibilities

- message **AssessAllocationPossibilities.Request**
- message **AssessAllocationPossibilities.Response**
- message **AssessAllocationPossibilities.Failure**

7.2.3 ReallocateAll

- message **ReallocateAll.Request**
- message **ReallocateAll.Response**
- message **ReallocateAll.Failure**

7.3 Type Reference

7.3.1 AssessAllocationPossibilities.Failure

empty **struct**

7.3.2 AssessAllocationPossibilities.Request

empty **struct**

7.3.3 AssessAllocationPossibilities.Response

struct

chipsRequired	int
---------------	-----

7.3.4 CardAllocation

struct

slot	optional int
inputInstance	optional int
outputInstance	optional int
enableDeinterlacer	optional bool
useCase	optional list of ResourceType
expectedInput	expectedInput.ExpectedInput

7.3.5 CardAllocationObj

CardAllocation

7.3.6 GetCardAllocations.Request

empty struct

7.3.7 GetCardAllocations.Response

struct

data	map from UUID to CardAllocationObj
info	GetCardAllocations.Response.info

7.3.8 GetCardAllocations.Response.info

struct

totalEntries	int
--------------	-----

7.3.9 ReallocateAll.Failure

empty struct

7.3.10 ReallocateAll.Request

empty struct

7.3.11 ReallocateAll.Response

struct

success

bool

7.3.12 ResourceType

enum

NOT_USE

RES_4K2K

FULL_HD

HD

SD

8 cardConfig (1.3)

8.1 Overview

Changelog

1.3

- **Changed**
 - nmosConfig from 1.0 to 1.1

1.2

- **Added**
 - nmosConfig 1.0
- **Removed**
 - nmosRegistryConfig 1.0

8.2 Command Reference

8.2.1 GetCardConfig

- message **GetCardConfig.Request**
- message **GetCardConfig.Response**
- message **GetCardConfig.Failure**

8.2.2 SetCardConfig

- message **SetCardConfig.Request**
- message **SetCardConfig.Response**
- message **SetCardConfig.Failure**

8.3 Type Reference

8.3.1 CardConfig

CardConfigStruct

8.3.2 CardConfigStruct

Card configuration parameters

struct

timeReference

timeMode.TimeMode

Time reference (PTP/NTP)

nmos

optional nmosConfig.NmosConfig

Parameters for accessing NMOS registry and Label Templates

8.3.3 GetCardConfig.Failure

empty **struct**

8.3.4 GetCardConfig.Request

empty **struct**

8.3.5 GetCardConfig.Response

struct

config **CardConfig**

8.3.6 SetCardConfig.Failure

empty **struct**

8.3.7 SetCardConfig.Request

struct

config **CardConfig**

8.3.8 SetCardConfig.Response

empty **struct**

9 cardStatus (1.9)

9.1 Overview

Changelog:

1.9

- **Changed**
 - nmosStatusTypes from 1.1 to 1.2

1.8

- **Changed**
 - Moved NmosStatus to nmosStatusTypes version 1.1

1.7

- **Added**
 - nodeApi map of interfaces to node API URIs
 - NmosStatus structure defined, and added to cardStatus structure.
 - NmosRegistryStatus put into NmosStatus.
- **Changed**
 - LinkModes from 1.2 to 1.3

1.6

- **Changed**
 - linkModes from 1.1 to 1.2

1.5

- **Changed**
 - sfpStatus from 1.4 to 1.5

9.2 Command Reference

9.2.1 GetCardStatus

- message **GetCardStatus.Request**
- message **GetCardStatus.Response**
- message **GetCardStatus.Failure**

9.3 Type Reference

9.3.1 CardStatus

struct

physicalPortStatus	optional PhysicalPortStatus
ptpLock	optional bool
nmosStatus	optional nmosStatusTypes.NmosStatus

qsfpStatus	optional map from	physicalIpPort.PortName	to	sfpStatus.SfpStatus
------------	-------------------	-------------------------	----	---------------------

9.3.2 DataRate

bigint

9.3.3 GetCardStatus.Failure

empty struct

9.3.4 GetCardStatus.Request

GetCardStatusRequest

9.3.5 GetCardStatus.Response

CardStatus

9.3.6 GetCardStatusRequest

struct

slot	int
------	-----

9.3.7 PhysicalPortStatus

struct

linkSpeed	map	from	physicalIpPort.PortName	to	linkModes.IpLinkSpeed
dataRate	map	from	physicalIpPort.PortName	to	DataRate
lldpNeighbor	map	from	physicalIpPort.PortName	to	lldp.LldpNeighborStruct

10 codedAudio (1.1)

10.1 Overview

```
# Changelog
## 1.1
- **Added**
  - `AC4` to `AudioCodec`
```

10.2 Type Reference

10.2.1 AudioCodec

enum

PCM

MPEG1L2

MPEG1L3

AAC_LC

AAC_HEv1

AAC_HEv2

AC3

EAC3

DOLBY_E

AC4

10.2.2 Hack

struct

profile

int

11 coderConfig (2.44)

11.1 Overview

Changelog:

2.44

- *Changed**
 - coderService from 2.41 to 2.42

2.43

- *Changed**
 - rawVideo from 1.4 to 1.5
 - coderService from 2.40 to 2.41
 - cardAllocation from 1.7 to 1.8

2.42

- *Changed**
 - coderService from 2.39 to 2.40
 - cardAllocation from 1.6 to 1.7

2.41

- *Changed**
 - coderService from 2.38 to 2.39

2.40

- *Changed**
 - coderService from 2.37 to 2.38

2.39

- **Changed**
 - coderService from 2.36 to 2.37
 - cardAllocation from 1.5 to 1.6

2.38

- **Changed**
 - coderService from 2.35 to 2.36

2.37

- **Changed**
 - coderService from 2.34 to 2.35

2.36

- **Changed**
 - coderService from 2.33 to 2.34

2.35

- **Changed**

- coderService from 2.32 to 2.33

2.34

- **Changed**

- coderService from 2.31 to 2.32

2.33

- **Changed**

- coderService from 2.30 to 2.31

2.32

- **Changed**

- imported esamConfig version 1.0
- added esamConfig to MultiServiceInfo
- coderService from 2.29 to 2.30

2.31

- **Changed**

- coderService from 2.28 to 2.29

2.30

- **Changed**

- coderService from 2.27 to 2.28

2.29

- **Changed**

- coderService from 2.26 to 2.27

2.28

- **Changed**

- added ready bool to StartupResponse
- coderService from 2.25 to 2.26

11.2 Type Reference

11.2.1 Foo

struct

id	UUID
service	coderService.CoderService
multiServiceInfo	optional MultiServiceInfo
cardAllocation	cardAllocation.CardAllocation

11.2.2 MessageType

enum

NEW
UPDATE
DELETE

11.2.3 MultiServiceInfo

struct

id	UUID
serviceIndex	int
profileMode	multiServiceProfileMode.ProfileMode
lowestFrameRateInMultiService	optional rawVideo.FrameRate
esamConfig	optional esamConfig.EsamConfig

12 coderService (2.42)

12.1 Overview

```
# Changelog
## 2.42
- **Added**
  - new optional `latencyOverride` member to the `CoderService` type.
- **Changed**
  - `audioProfile` from 2.5 to 2.6

## 2.41
- **Changed**
  - `videoProfile` from 2.10 to 2.11
  - `vancProfile` from 1.6 to 1.7

## 2.40
- **Changed**
  - `videoProfile` from 2.9 to 2.10

## 2.39
- **Changed**
  - Decoder IDCL migrations, in order to support downgrading to older versions
  - `colorComponent` from 1.0 to 1.1
  - `colorProfile` from 2.1 to 2.2
  - `vancProfile` from 1.5 to 1.6
  - `tsSource` from 1.4 to 1.5

## 2.38
- **Changed**
  - `audioProfile` from 2.4 to 2.5

## 2.37
- **Added**
  - New optional `pidSelection` member to `VancComponent`.
  - New list `pidList` member to `VancComponent`.
- **Changed**
  - `videoProfile` from 1.8 to 1.9

## 2.36
- **Changed**
  - `tsSource` from 1.3 to 1.4

## 2.35
- **Added**
  - `LR_CLfe` to `ChannelMapping` to match 3.1 channel mode use-case.
  - `identifier` optional field added to VancComponent struct, to be able to reference individual
- **Changed**
  - `tsSource` from 1.2 to 1.3
  - `ipStream` from 1.2 to 1.3
  - `ipService` from 1.1 to 1.2
  - `audioProfile` from 2.3 to 2.4
  - `vancProfile` from 1.4 to 1.5

## 2.34
- **Added**
  - `logoInsertion` 1.0
- **Changed**
  - EmbeddedAudio to use EmbeddedAudioChannels to enable custom source channels on encoders.
```

```
## 2.33
- **Changed**
  - `vancProfile` from 1.3 to 1.4
  - `EmbeddedAudio` `codec` made optional. Combine `codec` as empty optional with `passthrough`

## 2.32
- **Added**
  - `s302mPayload` to `TsAudioSource` in order to specify the content of a s302m PID

## 2.31
- **Changed**
  - `poolCtrlAllocs` from 1.0 to 1.1

## 2.30
- **Added**
  - `scte35Config` 1.0

- **Changed**
  - `idrOnScte35` optional bool changed to optional `Scte35Config`

## 2.29
- **Added**
  - Optional list of `AudioComponentPattern` added to `EmbeddedAudio` to be used by ABR JPEG XS
- **Changed**
  - `ipService` from 1.0 to 1.1
  - Moved the `ColorComponent` type to its own file.
  - `ipConnection` removed as it is no longer in use here.

## 2.28
- **Added**
  - Added ipService import which has the IpService type which supports input profile redundancy.
- **Changed**
  - `VideoOut` edited to fit the new input redundancy
  - `ipConnection` from 1.2 to 1.3

## 2.27
- **Added**
  - `latencyAdjustment` field added to AudioComponent for forwarding audio earlier than video.

## 2.26
- **Added**
  - `slateInsertion` 1.0
  - slateInsertion field added to CoderService for configuration of the Slate Insertion feature.

## 2.25
- **Changed**
  - `component` from 1.3 to 1.5
  - `tsSource` from 1.1 to 1.2

## 2.24
- **Changed**
  - `testGenComponent` from 1.0 to 1.1
  - `testGeneratorProfile` from 1.0 to 1.1

## 2.23
- **Added**
  - `sdiConnection` 1.0
  - `tsDestination` 1.0
  - `testGenComponent` 1.0
- **Removed**
```

- `SdiConnection` struct, moved to a separate file
- `SdiConnectors` enum, moved to a separate file
- `TestGenComponent` struct, moved to a separate file
- `TsDestination` struct, moved to a separate file

12.2 Command Reference

12.2.1 GetCoderServices

- message **GetCoderServices.Request**
- message **GetCoderServices.Response**
- message **GetCoderServices.Failure**

12.2.2 SetCoderServices

- message **SetCoderServices.Request**
- message **SetCoderServices.Response**
- message **SetCoderServices.Failure**

12.2.3 DeleteCoderServices

- message **DeleteCoderServices.Request**
- message **DeleteCoderServices.Response**
- message **DeleteCoderServices.Failure**

12.3 Type Reference

12.3.1 AudioComponent

struct

uid	int
source	AudioSource
numAuPerPes	optional int
lipSyncAdjustment	optional int
latencyAdjustment	optional int
levelAdjustment	optional int
loudness	optional Loudness
passthrough	bool
profile	AudioComponent.profile
destination	AudioOut

12.3.2 AudioComponent.profile

struct

id	optional UUID
cpy	optional audioProfile.AudioProfileStruct

12.3.3 AudioComponentPattern

struct

language	string
component	optional component.ComponentType
pid	optional int

12.3.4 AudioComponentSet

struct

main	AudioComponent
backup	optional AudioComponent

12.3.5 AudioOut

variant

embedded	EmbeddedAudio
ts	TsAudio

12.3.6 AudioSource

variant

embedded	EmbeddedAudio
ts	TsAudioSource

12.3.7 AudioType

enum

UNDEFINED
CLEAN_EFFECTS
HEARING_IMPAIRED
VISUALLY_IMPAIRED

12.3.8 ChannelMapping

enum

C
LR
LR_CLfe
LR_CLfe_LsRs
LR_LsRs_CLfe
LC_RLs_RsLfe
LLs_CRs_RLfe
LC_RsR_LsLfe

CL_RLs_RsLfe
 LR_CLfe_LsRs_LbRb
 LR_LsRs_CLfe_LbRb
 LC_RLs_RsLfe_LbRb
 LLs_CRs_RLfe_LbRb
 LC_RsR_LsLfe_LbRb
 CL_RLs_RsLfe_LbRb

12.3.9 CoderService

Configuration for an encoder or decoder service.

struct

label	string A label for the service.
enabled	bool True if the service should be enabled.
slot	int Which slot the service should run on (not applicable for Transcoders).
coderType	CoderType The type of coder the service is for.
video	VideoComponent Configuration of the input and output video for the coder chip.
audios	list of AudioComponentSet Configuration for the audio components.
output	OutputType The type of output that the service will produce.
color	optional colorComponent.ColorComponent Configuration for the color of the video.
vancs	list of VancComponent Configuration for the vanc components.
signalLoss	testGenerator.SignalLoss Configuration for the output when input signal is lost.
lock	LockComponent Configuration for how the service should be locked.
passthroughComponents	optional bool Set to true if all components should be passthrough. Only valid for transcoder.
additionalLatency	optional int Additional latency to be added to the entire service. Unit is 100ms. The optional value should be unset for products which do not support this feature.
latencyOverride	LatencyOverride Latency override for encoders. Use to override the video latency through the encoder. Auto finds a suitable value automatically, manual can be used to set a custom latency. optional testGenComponent.TestGenComponent

testGenerator	Contains optional structure with link to a test generator profile.
scte35	optional scte35Config.Scte35Config
autoSelectUniqueAudioSourcePid	optional bool Automatically select unique PIDs for audio components. PIDs are selected in order of increasing numerical PID value from the PMT such that the first audio component uses the lowest PID, second audio component the next to lowest PID and so forth. PMT updates with changes to PID lineup will trigger a reconfiguration of all audio components. If the number of audio components exceed the number of audio PIDs in the PMT the extra audio components will not run. Only valid for decoders.
slateInsertion	optional slateInsertion.SlateInsertion Enable the slate insertion feature. Only supported on encoder.
logoInsertion	optional logoInsertion.LogoInsertion Contains optional structure with configuration for logo insertion.

12.3.10 CoderServiceObj

CoderService

12.3.11 CoderType

The types of coders that a config can be made for.

enum

ENCODER
TRANSCODER
DECODER

12.3.12 DdppDecDrcMode

enum

NONE
LINE
RF

12.3.13 DdppDecoder

DdppDecoder.

@param ddppDecDrcMode Selects the mode for Dynamic Range Compression.
@param stereoDownmixMode Selects the mode for downmixing.

struct

ddppDecDrcMode **DdppDecDrcMode**
stereoDownmixMode **StereoDownmixMode**

12.3.14 DeleteCoderServices.Failure

empty **struct**

12.3.15 DeleteCoderServices.Request

struct

ids **list of UUID**

12.3.16 DeleteCoderServices.Response

empty **struct**

12.3.17 DolbyAudio

Dolby Audio.

@param dolbyEProgNum Selects a Dolby E program as source for the decoder and transcoder.
 @param ddppdec Specifies configuration for Dolby Digital and Dolby Digital Plus decoding.

struct

dolbyEProgNum **optional int**
 ddppdec **optional DdppDecoder**

12.3.18 EmbeddedAudio

Configuration for embedded audio.

struct

codec	optional codedAudio.AudioCodec Audio codec
channels	EmbeddedAudioChannels Embedded audio content
dolbyEProgNum	optional int Dolby E program number
channelMode	optional rawAudio.ChannelMode Channel mode – mono, stereo, 5.1, etc.
channelMapping	optional ChannelMapping Dictates how embedded audio channels are interpreted with respect to order.
audioEssenceId	optional UUID Audio essence UUID
componentPriorityList	optional list of AudioComponentPattern Optional prioritized list of input components from a transport stream.

12.3.19 EmbeddedAudioChannels

States where embedded audio content should be read from or written to in the SDI signal.

struct

channels

list of int

The source or destination channel(s) for the audio content

isCustom

bool

States whether a user specified list deviates from the standard set, i.e. non-adjacent channels.

12.3.20 Genlock**struct**

offset

int**12.3.21 GetCoderServices.Failure**empty **struct****12.3.22 GetCoderServices.Request**empty **struct****12.3.23 GetCoderServices.Response****struct**

data

map from **UUID** to **CoderServiceObj**

info

GetCoderServices.Response.info**12.3.24 GetCoderServices.Response.info****struct**

totalEntries

int**12.3.25 IpOut****string****12.3.26 LatencyOverride****variant**

auto

LatencyOverride.auto

manual

int**12.3.27 LatencyOverride.auto**empty **struct**

12.3.28 LockComponent

struct

source	optional bool
genlock	optional Genlock
videoAlignment	optional VideoAlignment

12.3.29 LockComponentEnum

enum

SOURCE
GENLOCK
VIDEOALIGNMENT
VIDEOALIGNMENT_AND_GENLOCK
NONE

12.3.30 Loudness

struct

longTermLoudnessTargetLevel	int
shortTermDeviation	optional int
peakLimit	optional int

12.3.31 OutputType

The types of output that a service can produce.

variant

ts	tsDestination.TsDestination
sdi	SdiOut
ip	IpOut

12.3.32 PidSelection

enum

ALL
ANY
LIST

12.3.33 SdiOut

empty struct

12.3.34 SetCoderServices.Failure

empty **struct**

12.3.35 SetCoderServices.Request

struct

data **map** from **UUID** to **CoderServiceObj**

12.3.36 SetCoderServices.Response

empty **struct**

12.3.37 StereoDownmixMode

enum

AUTO

LTRT

LORO

12.3.38 TsAudio

struct

language	string
pid	int
audioType	AudioType

12.3.39 TsAudioSource

Transport Stream Audio Source.

@param componentPriorityList A list of AudioComponentPattern objects describing audio components

@param dolbyAudio Specifies decode configuration for Dolby Digital, Dolby Digital Plus

@param s302mPayload Codec present in the s302m payload. Leave empty for transparent pas

struct

componentPriorityList	list of AudioComponentPattern
dolbyAudio	DolbyAudio
s302mPayload	optional codedAudio.AudioCodec

12.3.40 TsService

struct

pid	int
-----	------------

12.3.41 VancComponent

Configuration of VANC component

struct

uid	int Unique identifier for a component.
profile	VancComponent.profile Reference and copy to a VANC profile.
pidSelection	optional PidSelection Optional options for which PIDs to be selected
pidList	list of int List of PIDs to accompany PidSelection.LIST.
pid	int PID.
vancParams	vancProfile.VancParams Configuration of the type of ancillary data.
identifier	optional int Optional int to reference individual components.

12.3.42 VancComponent.profile

struct

id	UUID
cpy	optional vancProfile.VancProfileStruct

12.3.43 VideoAlignment

struct

index	int
alignmentMaster	optional poolCtrlAllocs.SlotFlow

12.3.44 VideoComponent

Video component of the coder service.

struct

source	VideoSource The input video
profile	VideoComponent.profile Video profile used by the coder service
destination	VideoOut The output video
timecodeSource	optional timecodeConfig.TimecodeSource Source of the timecode to be used for SEI messages (only set for encoders, but not for JPEG 2000 encoder)

12.3.45 VideoComponent.profile

struct

id	UUID
cpy	optional videoProfile.VideoProfileStruct

12.3.46 VideoOut

variant

sdi	sdiConnection.SdiConnection
ts	TsService
ip	ipService.IpService

12.3.47 VideoSource

```
Configuration type for the video source.  
@param sdi configuration for SDI video source.  
@param ts configuration for TS video source.  
@param ip configuration for IP video source.
```

variant

sdi	sdiConnection.SdiConnection
ts	tsSource.TsSource
ip	ipService.IpService

13 coderTypes (1.0)

13.1 Type Reference

13.1.1 Range

struct

min	int
max	int

14 colorComponent (1.1)

14.1 Overview

```
# Changelog
## 1.1
- **Changed**
  `colorProfile` from 2.1 to 2.2

## 1.0
- **Added**
  - Moved the ColorComponent type from coderService
```

14.2 Type Reference

14.2.1 ColorComponent

Configuration for a single color profile.

struct

profile

ColorComponent.profile

The color profile with a UUID and a copy of the color profile struct

14.2.2 ColorComponent.profile

struct

id

UUID

cpy

optional colorProfile.ColorProfileStruct

15 colorProfile (2.2)

15.1 Command Reference

15.1.1 GetColorProfiles

- message **GetColorProfiles.Request**
- message **GetColorProfiles.Response**
- message **GetColorProfiles.Failure**

15.1.2 SetColorProfiles

- message **SetColorProfiles.Request**
- message **SetColorProfiles.Response**
- message **SetColorProfiles.Failure**

15.1.3 DeleteColorProfiles

- message **DeleteColorProfiles.Request**
- message **DeleteColorProfiles.Response**
- message **DeleteColorProfiles.Failure**

15.2 Type Reference

15.2.1 ColorProfile

ColorProfileStruct

15.2.2 ColorProfileStruct

struct

label	string
colorType	ColorType
vui	videoUsabilityInfo.VuiParameters
followInput	bool

15.2.3 ColorType

```
# Changelog
## 2.2
- **Changed**
  Made use of common definition of VUI parameters instead of local ones
  Added BT_601 to `ColorSpace`
  Added BT_601 to `TransferFunc`
```

enum

```
OFF
HLG_10
HDR_10
PQ_10
```

SDR	
SDR_WCG	
HLG_10_ATF	
Manual	

15.2.4 DeleteColorProfiles.Failure

empty **struct**

15.2.5 DeleteColorProfiles.Request

struct

ids **list** of **UUID**

15.2.6 DeleteColorProfiles.Response

empty **struct**

15.2.7 GetColorProfiles.Failure

empty **struct**

15.2.8 GetColorProfiles.Request

empty **struct**

15.2.9 GetColorProfiles.Response

struct

data	map from UUID to ColorProfile
info	GetColorProfiles.Response.info

15.2.10 GetColorProfiles.Response.info

struct

totalEntries **int**

15.2.11 SetColorProfiles.Failure

empty **struct**

15.2.12 SetColorProfiles.Request

struct

data

map from UUID to ColorProfile

15.2.13 SetColorProfiles.Response

empty struct

16 commonTypes (1.0)

16.1 Type Reference

16.1.1 NmosVersionNumber

Generic version number type for NMOS APIs

enum

V1_2

V1_3

17 component (1.5)

17.1 Overview

Changelog:

1.5

- Added
 - SCTE27_SUB to ComponentType enum and respective mappings.

17.2 Type Reference

17.2.1 AnyComponent

empty **struct**

17.2.2 ComponentGroup

ComponentGroup: Grouping of ComponentType, used for filtering

enum

UNDEFINED
AUDIO
VIDEO
SUBTITLE
DATA

17.2.3 ComponentLanguage

string

17.2.4 ComponentSelector

Component selector.

@param pid	Selects a component using a specific (source) PID.
@param language	Selects a component by a three character language string.
@param type	Selects a component by its type.
@param group	Selects a component by group membership.
@param any	Selects any component.

variant

pid	Pid
language	ComponentLanguage
componentType	ComponentType
componentGroup	ComponentGroup
any	AnyComponent

17.2.5 ComponentType

A ComponentType is typically identified by a stream type (see ISO 13818-1 Table 2-29), often distinguished with another header field such as a descriptor tag.

enum

MPEG_VIDEO

MPEG_AUDIO

TTX

DVBSUB

ECM

PCR

PRIVATE

AC_3

H264

VBI

AAC

VC_1

SCS_EMM

DPI_CUE

EAC3

AIT

HBBTV_CAROUSEL

DATA_CAROUSEL

H265

S302M

JPEG2000

S2038

DVBTTMLSUB

JPEG_XS_VIDEO

SCTE27_SUB

17.2.6 Pid

int

17.2.7 SubtitleFormat

enum

DVB

DVB_HARD_OF_HEARING

EBU

EBU_HARD_OF_HEARING

18 de jitterTypes (1.1)

18.1 Overview

Changelog:

1.1

- Added
 - PCR value added to De jitterType enum
 - PCR added to De jitterType2str

18.2 Type Reference

18.2.1 De jitterType

enum

OFF

RTP_2022_6

PCR

19 dpi (1.0)

19.1 Type Reference

19.1.1 Dpi

Configuration of the DPI data

struct

source	DpiSource Source of the DPI data
heartbeat	bool Whether heartbeat is on or off
heartbeatInterval	int Interval between each heartbeat
ptsOffset	int SCTE-35 PTS adjustment
index	int SCTE 104 DPI PID index

19.1.2 DpiSettings

Mapping an identifier to the DPI data

struct

identifier	optional int The identifier for mapping SCTE104 messages to the service.
dpiData	Dpi Configuration of the DPI data

19.1.3 DpiSource

Source of the DPI data

enum

ANC	
TCP	

20 dpiStatus (1.0)

20.1 Type Reference

20.1.1 DpiCommand

DPI Command info

struct

commandType	int Last SCTE command type.
descriptorIds	list of int List of segmentation descriptor IDs.

20.1.2 DpiInput

DPI input status.

struct

sourceIdentifier	optional int identifies the source of the input. Encoder: ANC is represented by -1, TCP input is represented by its identifier.
received	bool Received DPI input.
invalidPackets	int Input contained invalid packets.
pidIndexes	list of int
timeSinceLastReceived	int
lastGeneratedCommand	optional DpiCommand

20.1.3 DpiStatus

DPI status.

struct

heartbeat	bool Specifies whether heartbeat signal is on or off.
enabled	bool Specifies whether DPI is enabled.
running	bool Specifies whether DPI encoding is running without errors.
pidIndex	int SCTE PID index.
timeSinceLastReceived	int Seconds since last received DPI packet.
timeSinceLastGenerated	int Seconds since last generated DPI packet.
lastCommandType	int Type of last received SCTE command.

`discardedPackets`**int**

Number of discarded packets.

21 esamConfig (1.0)

21.1 Type Reference

21.1.1 EsamConfig

struct

poisUri **string**

21.1.2 EsamServiceConfig

Multiservice ESAM configuration.

struct

acquisitionPointId **string**
The acquisition point ID that will be sent to the POIS server.

22 esamStatus (1.0)

22.1 Type Reference

22.1.1 EsamStatus

ESAM status.

struct

poisConnectionStatus

PoisConnectionStatus

An enum to represent the connection status with the POIS server.

22.1.2 PoisConnectionStatus

UNKNOWN means no communication with the server yet

enum

CONNECTED

NO_CONNECTION

UNKNOWN

23 expectedInput (1.5)

23.1 Overview

Changelog

1.5

- **Changed**
 - `videoProfile` from 2.10 to 2.11
 - `rawVideo` from 1.4 to 1.5

1.4

- **Changed**
 - `videoProfile` from 2.9 to 2.10

1.3

- **Changed**
 - `videoProfile` from 2.8 to 2.9

1.2

- **Added**
 - Added ExpectedTsInput struct
 - Added ExpectedRawInput variant
 - New fields in ExpectedInput:
 - ts: ExpectedTsInput,
 - raw: ExpectedRawInput
- **Changed**
 - ExpectedInput changed from a struct to a variant of either ExpectedTsInput or ExpectedRawInput

23.2 Type Reference

23.2.1 ExpectedInput

Expected input format.

variant

ts

ExpectedTsInput

the expected input format for a TS video source.

raw

ExpectedRawInput

the expected input format for a RAW video source.

23.2.2 ExpectedRawInput

Expected raw input.

struct

rawVideo.HorizontalResolution

hResolution	the expected horizontal resolution of the input.
vResolution	rawVideo.VerticalResolution the expected vertical resolution of the input.
scan	videoProfile.ScanningMode the expected scanning mode of the input.
fps	rawVideo.FrameRate the expected frame rate per second of the input.

23.2.3 ExpectedTsInput

Expected ts input.

struct

interlaced	bool Enables handling of interlaced inputs. If set it may consume more resources.
codec	optional videoProfile.VideoCodec Empty optional represents Any codec (in set MPEG2 AVC HEVC)
allow4k	bool Enables handling of 4k inputs. If set it will consume more resources.
allow10bit	bool Enables handling of 10bit inputs. If set it will consume more resources.

24 firewallTypes (1.0)

24.1 Type Reference

24.1.1 IPAddress

string

24.1.2 PaginatedRequest

Instructs the server to return only a subset of the data found on the server

@param pageSize number of entries returned per request

@param pageNumber what page of the set to returned, determined by the slice [pageSize*pageNumber

struct

pageSize	int
pageNumber	int

24.1.3 PaginationInfo

struct

totalEntries	int
--------------	-----

the total number of entries on the server, before matching

24.1.4 SocketAddress

Tuple with IP address and port

struct

address	IPAddress
port	int

IP address, either IPv4 or IPv6

24.1.5 UUID

string

25 imageUpload (1.0)

25.1 Overview

```
# Changelog

## 1.0
- **Added**
  - First version of module
```

25.2 Command Reference

25.2.1 GetImages

- message **GetImages.Request**
- message **GetImages.Response**

25.2.2 SetImage

Must be sent after an image has been uploaded. It will make sure that image is stored permanently and is available for selection.

- message **SetImage.Request**
- message **SetImage.Response**

25.2.3 DeleteImage

- message **DeleteImage.Request**
- message **DeleteImage.Response**

25.3 Type Reference

25.3.1 DeleteImage.Request

struct

id **string**

25.3.2 DeleteImage.Response

empty **struct**

25.3.3 GetImages.Request

empty **struct**

25.3.4 GetImages.Response

struct

images **list of ImageInfo**

25.3.5 ImageInfo

Information about an image.

struct

id	string a unique identifier for the image.
fileName	string the image file name.

25.3.6 SetImage.Request

struct

fileName	string
----------	---------------

25.3.7 SetImage.Response

struct

fileId	string A unique ID for the uploaded image. It is important to be aware that the ID is based on the content of the file. So if identical files (with different file names) are uploaded they will overwrite eachother.
--------	---

26 inputRedundancyStatus (1.3)

26.1 Overview

```
# Changelog

## 1.3
- **Changed**
  - `psiStatus` from 1.5 to 1.6

## 1.2
- **Changed**
  - `psiStatus` from 1.4 to 1.5
```

26.2 Type Reference

26.2.1 HotStandbyStatus

Hot Standby status for input redundancy.

struct

identicalSourcesStatus **optional** **IdenticalSourcesStatus**

26.2.2 IdenticalSourcesStatus

Identical sources status for input redundancy.

struct

measuredOffset **int**
Offset between two identical sources in ms.

26.2.3 SourceStatus

Source status for input redundancy.

struct

components	psiStatus.PsiStatus Components present in the given source.
present	bool True if the source is present, false otherwise.
active	bool True if the source is the active one in a input redundancy set-up.

26.2.4 TsInputRedundancyStatus

TS input redundancy status.

struct

sourceStatus	map from streamSource.StreamSource to SourceStatus Status for the given stream source.
hotStandbyStatus	HotStandbyStatus

27 ipConnection (1.5)

27.1 Overview

Changelog:

1.5

- Changed
 - ipStream from 1.2 to 1.3

1.4

- Changed
 - deJitterTypes from 1.0 to 1.1

1.3

- Added
 - MPEG_TS value added to IpStandard enum

1.2

- Added
 - nmosEnable field added to IpConnection, with a default value of false

27.2 Command Reference

27.2.1 GetIpConnections

- message **GetIpConnections.Request**
- message **GetIpConnections.Response**
- message **GetIpConnections.Failure**

27.2.2 SetIpConnections

- message **SetIpConnections.Request**
- message **SetIpConnections.Response**
- message **SetIpConnections.Failure**

27.2.3 DeletelpConnections

- message **DeletelpConnections.Request**
- message **DeletelpConnections.Response**
- message **DeletelpConnections.Failure**

27.3 Type Reference

27.3.1 DeJitter

struct

mode	deJitterTypes.DeJitterType
bufferSize	int

27.3.2 DeleteIpConnections.Failure

empty **struct**

27.3.3 DeleteIpConnections.Request

struct

ids **list of UUID**

27.3.4 DeleteIpConnections.Response

empty **struct**

27.3.5 GetIpConnections.Failure

empty **struct**

27.3.6 GetIpConnections.Request

empty **struct**

27.3.7 GetIpConnections.Response

struct

data	map from UUID to IpConnectionObj
info	GetIpConnections.Response.info

27.3.8 GetIpConnections.Response.info

struct

totalEntries **int**

27.3.9 IpConnection

Group of settings for the IP inputs/outputs of a single service.

struct

label	string Label for the entire Group.
connection	ipStream.Connection Connections settings like addresses, ports, etc.
de jitter	optional De jitter De jitter for the input of IP 2110 and IP 2022 encoders.
standard	IpStandard SMPTE standard that is used for this group.

bool

nmosEnable

for allowing NMOS system to update the input/output profiles

27.3.10 IpConnectionObj

IpConnection

27.3.11 IpStandard

enum

SMPTE_2022

SMPTE_2110

MPEG_TS

27.3.12 SetIpConnections.Failure

empty struct

27.3.13 SetIpConnections.Request

struct

data

map from UUID to IpConnectionObj

27.3.14 SetIpConnections.Response

empty struct

28 ipInterface (1.6)

28.1 Overview

```
# Changelog

## 1.6
- **Changed**
`linkModes` import from 1.2 to 1.3

## 1.5
- **Added**
`ipFecMode` added to IpInterfaceStruct

## 1.4
- **Changed**
`lldp` import from 1.0 to 1.1
```

28.2 Command Reference

28.2.1 GetIpInterfaces

- message **GetIpInterfaces.Request**
- message **GetIpInterfaces.Response**
- message **GetIpInterfaces.Failure**

28.2.2 SetIpInterfaces

Interface uuids are deterministically calculated The uuids used in the RPC Request are ignored

- message **SetIpInterfaces.Request**
- message **SetIpInterfaces.Response**
- message **SetIpInterfaces.Failure**

28.2.3 DeleteIpInterfaces

- message **DeleteIpInterfaces.Request**
- message **DeleteIpInterfaces.Response**
- message **DeleteIpInterfaces.Failure**

28.3 Type Reference

28.3.1 DeleteIpInterfaces.Failure

empty **struct**

28.3.2 DeleteIpInterfaces.Request

struct

ids **list** of **UUID**

28.3.3 DeleteIpInterfaces.Response

empty **struct**

28.3.4 GetIpInterfaces.Failure

empty **struct**

28.3.5 GetIpInterfaces.Request

empty **struct**

28.3.6 GetIpInterfaces.Response

struct

data **map** from **UUID** to **IpInterface**

28.3.7 IpInterface

IpInterfaceStruct

28.3.8 IpInterfaceConfig

struct

ipAddress	firewallTypes.IpAddress
gateway	firewallTypes.IpAddress
netmask	int

28.3.9 IpInterfaceStruct

struct

ipLink	linkModes.IpLinkMode
ipFecMode	optional linkModes.IpFecMode
interfaces	IpInterfaceVariant

28.3.10 IpInterfaceVariant

variant

single	LogicalInterface
quad	IpInterfaceVariant.quad

28.3.11 IpInterfaceVariant.quad

struct

first **LogicalInterface**

second	LogicalInterface
third	LogicalInterface
fourth	LogicalInterface

28.3.12 LogicalInterface

struct

label	string
enabled	bool
physicalPort	physicalIpPort.PortName
ipv4	optional IpInterfaceConfig
ipv6	optional IpInterfaceConfig
lldpMode	optional Ildp.LldpMode

28.3.13 SetIpInterfaces.Failure

empty struct

28.3.14 SetIpInterfaces.Request

struct

data	map from UUID to IpInterface
------	------------------------------

28.3.15 SetIpInterfaces.Response

empty struct

29 ipService (1.2)

29.1 Overview

```
# Changelog
## 1.2
- **Changed**
  - `ipConnection` from 1.4 to 1.5
  - `ipStream` from 1.2 to 1.3

## 1.1
- **Changed**
  - `ipConnection` from 1.3 to 1.4

## 1.0
- **Added**
  - New type `IpServices` added, containing multiple IP profiles and switching mode in order to
  - Moved the IpProfile type from coderService
```

29.2 Type Reference

29.2.1 IpProfile

Configuration for a single IP video source.

struct

id	UUID UUID of the IpConnection profile containing the configuration.
cpy	optional ipConnection.IpConnection A copy of the content of the IpConnection profile.

29.2.2 IpService

Configuration for one or more IP video sources.

struct

profiles	list of IpProfile
profileRedundancy	optional ipStream.NearSeamless
sdiType	sdiType.SdiType

30 ipStatus (1.7)

30.1 Overview

Changelog:

1.7

- Changed
 - sfpStatus from 1.4 to 1.5

30.2 Type Reference

30.2.1 InstanceId

int

30.2.2 IpStatus

struct

id	InstanceId
tico	optional bool
sfp	optional sfpStatus.SfpStatus
ipStreamStatuses	optional list of ipStreamStatus.IpStreamStatus
alternateInputStatus	optional alternateInputStatus.AlternateInputStatus

31 ipStream (1.3)

31.1 Overview

```
# Changelog
## 1.3
- **Changed**
- `st2110Types` from 1.0 to 1.2.
```

31.2 Type Reference

31.2.1 AlternateInput

IP config for a service using alternative input.

struct

path1	IpStreamGroup Config for path one.
path2	IpStreamGroup Config for path two.
bufferSize	int Input buffer size in milliseconds.
seamlessType	SeamlessType Config for the different types of alternative input.

31.2.2 Connection

A redundant Ip Connection.

variant

direct	IpStreamGroup No redundancy.
alternateInput	AlternateInput Seamless or alternative input.

31.2.3 EssenceProperties

Essence-specific properties of a 2110 essence.

variant

audio	St2110AudioProperties Properties of an audio essence.
-------	---

31.2.4 IpStream

This struct stores the address and other configurable options of an input or output stream on an IP network. On 2110 products it represents an essence.

struct

label	string A descriptive label or name for this stream.
address	firewallTypes.SocketAddress The IP address of this stream.
igmpv3Source	optional firewallTypes.IpAddress IP address of the IGMP source used for source filtering.
payloadId	int Id of the essence.
st2110Info	optional St2110Info Information which is only available for 2110 applications.
id	UUID Unique id within an IpStreamGroup. Redundant streams (in different groups) have the same id.

31.2.5 IpStreamGroup

A group of IpStreams that share the same physical port.

struct

bufferSize	optional int Size of buffer used when synchronizing essences.
source	list of IpStream The list of IpStreams in the group. Shall contain a single entry for 2022, but may contain multiple for 2110.
physicalPortId	physicalIpPort.PortName The id of the physical port on which the IpStreams are sent/received.

31.2.6 NearSeamless

Config for near seamless mode.

struct

switchingMode	SwitchingMode The mode of switching.
timeoutMainToSpare	int Timeout before switching to spare in milliseconds.
timeoutSpareToMain	int Timeout before switching to main in milliseconds.

31.2.7 SeamlessType

Config for seamless.

variant

smpte2022_7	SeamlessType.smpte2022_7 Config for SMPTE 2022-7 seamless.
nearSeamless	NearSeamless Config for near seamless mode.

31.2.8 SeamlessType.smpte2022_7

empty **struct**

31.2.9 St2110AudioProperties

Properties specific to a 2110 audio essence. Either -30 or -31.

struct

totalChannels	int The number of audio channels in the essence.
packetTime	st2110Types.St2110AudioPacketTime Audio packet time.

31.2.10 St2110Info

Information regarding any kind of 2110 essence.

struct

standard	st2110Types.St2110Standard The standard which the essence content follows.
properties	EssenceProperties Properties specific to a type of essence.

31.2.11 SwitchingMode

The switching modes that can be used with alternative input.

enum

PREFER_PATH1
PREFER_PATH2
FLOATING
FORCE_PATH1
FORCE_PATH2

32 ipStreamStatus (1.1)

32.1 Type Reference

32.1.1 IpStreamStatus

Represents the status of an IpStream.

struct

id	UUID The id of the IpStream this status represents.
bitrate	float The bitrate of the IpStream.
networkProtocol	NetworkProtocol Protocol used for seamless input.
sequenceErrors	int Number of gaps in RTP sequence numbers.
rtpSsrc	int Synchronization Source that uniquely identifies the source of a stream. Ssrc's within an RTP session will be unique.
minDelay	int Minimum network latency. [ms]
maxDelay	int Maximum network latency. [ms]

32.1.2 NetworkProtocol

Possible protocols of a seamless input.

enum

RTP	
SequencedUDP	
UDP	
Unknown	

33 ipStreamUtils (0.0)

34 linkModes (1.3)

34.1 Type Reference

34.1.1 IpFecMode

enum

NO_FEC

RS_FEC

FC_FEC

34.1.2 IpLinkMode

enum

IP_10G

IP_25G_COPPER

IP_25G_OPTICAL

IP_40G

IP_4x10G

34.1.3 IpLinkSpeed

enum

IP_10G

IP_25G

IP_40G

IP_4x10G

NO_LINK

35 lldp (1.1)

35.1 Overview

```
# Changelog
```

```
## 1.1
```

```
- **Changed**
```

```
- Extracted the structure inside of the `LldpNeighbor` object definition into a new type cal
```

35.2 Type Reference

35.2.1 LldpMedInventory

LLDP-MED Inventory. Appear specific implementation in parentheses.

struct

hwRev	string Hardware Revision
swRev	string Software Revision (Software Version)
fwRev	string Firmware Revision (FPGA Version)
serial	string Serial Number
manufacturer	string Manufacturer ("Appear")
model	string Model ("X10" or "X20")
assetId	string Asset ID (Hardware ID)

35.2.2 LldpMode

Choose either private or public configuration of LLDP frames.

enum

PUBLIC	Only mandatory TLVs and port detail TLVs.
PRIVATE	LLDP-MED inventory in addition to the PUBLIC TLVs.

35.2.3 LldpNeighbor

LldpNeighborStruct

35.2.4 LldpNeighborStruct

The optional variables are only present if `GetLldpNeighborRequest.details` is set to true in the rpc request. The following descriptions are the Appear specific implementations of each LLDP TLV. If neighbors from other manufacturers are detected, TLV values might differ. The `LldpMode` is specified in

parentheses for variables that have different implementations based on the LldpMode set in the neighbor.

struct

chassisId	string MAC address of CTRL on MMI 1, fallback to CTRL on MMI 2 (PRIVATE) or "N/A" (PUBLIC).
mgmtIp	list of string IPv4 address of CTRL on MMI 1 and/or CTRL on MMI 2. Also IPv6 if enabled on the neighbor.
sysDescr	optional string Desktop Heading (PRIVATE) or "Unknown" (PUBLIC)
sysName	optional string Hostname (PRIVATE) or Interface.Port (PUBLIC)
portId	string MAC address of local port
portDescr	optional string Slot, Interface: Port-label
ttl	optional string Time-To-Live, how long an LLDP packet received from that neighbor should be considered valid.
lldpMedInventory	optional LldpMedInventory LLDP-MED Inventory

36 lockStatus (1.4)

36.1 Overview

Changelog:

1.4

- **Changed**
 - LockStatus is now a struct instead of a variant
 - LockStatus struct takes both genLockStatus and videoAlignmentStatus as optional

1.3

- **Changed**
 - poolCtrlAllocs from 1.0 to 1.1
- **Added**
 - flow from 1.0 to 1.1

36.2 Type Reference

36.2.1 GenlockStatus

struct

locked	bool
offset	int
validMode	bool
localReference	ReferenceLockStatus
remoteReference	ReferenceLockStatus

36.2.2 LockStatus

struct

genlockStatus	optional GenlockStatus
videoAlignmentStatus	optional VideoAlignmentStatus

36.2.3 ReferenceLockStatus

enum

UNLOCKED
LOCKED
HOLDOVER

36.2.4 VideoAlignmentStatus

struct

locked	bool
offset	int
group	int
alignmentMaster	poolCtrlAllocs.SlotFlow
outOfAlignment	bool
flow	int

37 logoInsertion (1.0)

37.1 Overview

```
# Changelog

## 1.0
- **Added**
  - First version of module
```

37.2 Type Reference

37.2.1 LogoInsertion

Configure logo insertion.

struct

id

string

A unique identifier for the logo. This is the value returned by the imageUpload.SetImage RPC.

xOffset

int

X offset from the top left corner in pixels.

yOffset

int

Y offset from the top left corner in pixels.

38 movedService (1.0)

38.1 Overview

```
# Changelog

## 1.0
- **Added**
  - added MovedServiceStruct and Obj
  - added slot type
```

38.2 Type Reference

38.2.1 MovedService

MovedServiceStruct

38.2.2 MovedServiceStruct

Struct that stores where services are being moved during service redundancy switch. For example in N+M redundancy mode

struct

serviceId	UUID Id of the service which will be moved
fromSlot	Slot From which slot the service will be moved
toSlot	optional Slot To which slot the service will be moved

38.2.3 Slot

Slot used by the service in the MovedServiceStruct

int

39 multiService (2.14)

39.1 Overview

Changelog

2.14

- **Changed**
 - coderService from 2.41 to 2.42

2.13

- **Changed**
 - coderService from 2.40 to 2.41

2.12

- **Changed**
 - coderService from 2.39 to 2.40

2.11

- **Changed**
 - coderService from 2.38 to 2.39
 - colorComponent from 1.0 to 1.1
 - tsSource from 1.4 to 1.5

2.10

- **Changed**
 - added list of vanc components to MultiService
 - coderService from 2.37 to 2.38

2.9

- **Changed**
 - coderService from 2.36 to 2.37

2.8

- **Changed**
 - tsSource from 1.3 to 1.4

2.7

- **Changed**
 - coderService from 2.34 to 2.35
 - tsSource from 1.2 to 1.3
 - ipService from 1.1 to 1.2

2.6

- **Added** -logoInsertion 1.0
- **Changed**

- coderService from 2.33 to 2.34

2.5

- **Changed**

- coderService from 2.32 to 2.33

2.4

- **Changed**

- coderService from 2.31 to 2.32

2.3

- **Changed**

- coderService from 2.30 to 2.31

2.2

- **Added**

- Scte35Config 1.0

- **Changed**

- idrOnScte35 bool is now moved from MultiServiceOptions to Scte35Options in scte35Config 1.0
- scte35 contains Scte35Options instead of idrOnmScte35 bool
- coderService from 2.29 to 2.30

2.1

- **Added**

- IpSource struct containing an IpService and a list of slots in use

- **Changed**

- coderService from 2.28 to 2.29
- ipService from 1.0 to 1.1
- added colorComponent import and colorProfile to MultiService
- replaced IpService as ip variant in MsSource with newly added IpSource

2.0

- **Changed**

- coderService from 2.27 to 2.28
- added ipService as a possible multiservice source
- moved audioOptions to MultiServiceOptions
- replaced audioOptions on ServiceOptions with audioSubscriptions to refer to shared audioOptions
- these changes are not backwards compatible

1.22

- **Added**

- latencyAdjustment field added to AudioOptions struct for forwarding audio earlier than video.

- **Changed**
 - coderService from 2.26 to 2.27

1.21

- **Changed**
 - coderService from 2.25 to 2.26

39.2 Command Reference

39.2.1 SetMultiServices

- message **SetMultiServices.Request**
- message **SetMultiServices.Response**

39.2.2 GetMultiServices

- message **GetMultiServices.Request**
- message **GetMultiServices.Response**

39.2.3 DeleteMultiServices

- message **DeleteMultiServices.Request**
- message **DeleteMultiServices.Response**

39.3 Type Reference

39.3.1 AudioOptions

Audio options for a given Transcoder or ABR encoder audio track.

struct

source	coderService.AudioSource Source configuration for a track.
numAuPerPes	optional int Number of access units per PES, commonly referred to as PES alignment.
lipSyncAdjustment	optional int Lip sync adjustment
latencyAdjustment	optional int Adjust latency of audio relative to video.
levelAdjustment	optional int Level adjustment.
loudness	optional coderService.Loudness Loudness configuration.
destination	optional coderService.TsAudio Destination configuration for an ABR encoder.
passthrough	optional bool Flag based on the presence of an audio profile in the multi service profile for a given audio track; missing profiles indicates passthrough (true) here. ABR encoder only.

39.3.2 DeleteMultiServices.Request

struct

ids **list of UUID**

39.3.3 DeleteMultiServices.Response

empty **struct**

39.3.4 GetMultiServices.Request

empty **struct**

39.3.5 GetMultiServices.Response

struct

data **map from UUID to MultiServiceObj**
 info **GetMultiServices.Response.info**

39.3.6 GetMultiServices.Response.info

struct

totalEntries **int**

39.3.7 IpSource

Configuration for IP sources.

struct

ipService **ipService.IpService**
 Configuration for one or more IP video sources.

slot **list of int**
 A list of slots utilizing the source.

39.3.8 MsSource

Multi Service Source. Configuration of the source of the Multi Service. Can be either ts, sdi or ip

variant

ts **tsSource.TsSource**
 Configuration of the TS source. Transcoder only

sdi **SdiSource**
 Configuration of the SDI source. ABR Encoder only

ip **IpSource**
 Configuration of the IP source.

39.3.9 MultiService

Configuration for a Transcoder or ABR Encoder multi service.

struct

label	string A label for the service.
enabled	bool True if the service should be enabled.
source	MsSource Configuration of the source.
profile	MultiService.profile The profile that describes the Multi Service
options	MultiServiceOptions Configuration of components and IDR on SCTE35.
signalLoss	optional testGenerator.SignalLoss Configuration for the output when input signal is lost. ABR Encoder only
testPatternGenerator	optional testGenComponent.TestGenComponent Contains optional structure with link to a test generator profile. ABR Encoder only.
colorProfile	optional colorComponent.ColorComponent Contains optional structure with link to a color profile.
vancs	list of coderService.VancComponent A list of vanc components. SDI ABR Encoder only.

39.3.10 MultiService.profile

struct

id	UUID
----	-------------

39.3.11 MultiServiceObj

MultiService

39.3.12 MultiServiceOptions

Options for a given Transcoder or ABR encoder service.

struct

scte35	scte35Config.Scte35Config Configuration of SCTE 35.
audioOptions	map from int to AudioOptions Configuration of audio coding instances.
serviceOptions	map from int to ServiceOptions Configuration of component services.

39.3.13 SdiSource

Configuration for an ABR encoder.

struct

sdiConnection	sdiConnection.SdiConnection Configuration of the SDI connection of the source
slot	int Which slot the source comes from.

39.3.14 ServiceOptions

- Service options for a specific Transcoder or ABR encoder service.
-

struct

videoOptions	optional VideoOptions Video options (unused). •
audioSubscriptions	list of int References to the audio tracks selected for this service. These identifiers must exist as keys in the map of AudioOptions. •
passthroughComponents	optional bool On transcoder only, setting this flag to true passes through all whitelisted transport stream components that are not transcoded by the service.

39.3.15 SetMultiServices.Request

struct

data	map from UUID to MultiServiceObj
------	---

39.3.16 SetMultiServices.Response

empty **struct**

39.3.17 VideoOptions

struct

logoInsertion	optional logoInsertion.LogoInsertion Contains optional structure with configuration for logo insertion.
---------------	---

40 multiServiceProfile (2.6)

40.1 Overview

Changelog

2.6

- **Changed**
 - audioProfile from 2.5 to 2.6

2.5

- **Changed**
 - videoProfile from 2.10 to 2.11

2.5

- **Changed**
 - videoProfile from 2.10 to 2.11

2.4

- **Changed**
 - videoProfile from 2.9 to 2.10
 - expectedInput from 1.3 to 1.4

2.3

- **Changed**
 - audioProfile from 2.4 to 2.5

2.2

- **Changed**
 - videoProfile from 2.8 to 2.9
 - expectedInput from 1.2 to 1.3

2.1

- **Changed**
 - audioProfile from 2.3 to 2.4

2.0

- **Changed**
 - moved audios to MultiServiceProfile
 - added audioSubscriptions to SubService to refer to shared audios
 - these changes are not backwards compatible

1.10

- **Changed**
- AudioTrack.profile field is changed from AudioProfileStruct to optional.
- expectedInput from 1.1 to 1.2

40.2 Command Reference

40.2.1 SetMultiServiceProfiles

- message **SetMultiServiceProfiles.Request**
- message **SetMultiServiceProfiles.Response**

40.2.2 GetMultiServiceProfiles

- message **GetMultiServiceProfiles.Request**
- message **GetMultiServiceProfiles.Response**

40.2.3 DeleteMultiServiceProfiles

- message **DeleteMultiServiceProfiles.Request**
- message **DeleteMultiServiceProfiles.Response**

40.3 Type Reference

40.3.1 AudioTrack

Audio Track configuration parameters.

@param uid An audio track's unique id. Values are not required to be contiguous.

@param profile A profile describing an audio encoding process. Empty optional signals passthrough

struct

uid	int
profile	optional audioProfile.AudioProfileStruct

40.3.2 DeleteMultiServiceProfiles.Request

struct

ids	list of UUID
-----	--------------

40.3.3 DeleteMultiServiceProfiles.Response

empty **struct**

40.3.4 GetMultiServiceProfiles.Request

empty **struct**

40.3.5 GetMultiServiceProfiles.Response

struct

data	map from UUID to MultiServiceProfileObj
info	GetMultiServiceProfiles.Response.info

40.3.6 GetMultiServiceProfiles.Response.info

struct

totalEntries **int**

40.3.7 MultiServiceProfile

Multiservice configuration parameters.
 @param profileName An text indentifier for the profile.
 @param profileMode Flag for broadcast or ABR profile modes.
 @param input Describes the expected input format of the video signal.
 @param gopSize Describes the distances in frames between two keyframes.
 @param aspectRatio Describes the ratio of width to height of the video images.
 @param frameRateDomain Describes the restricted set of frame rates these services should support
 @param audios List of audio configurations for this profile. Each AudioTrack can be used in mult
 @param subServices List of subservice definitions.

struct

profileName	string
profileMode	multiServiceProfileMode.ProfileMode
input	expectedInput.ExpectedInput
gopSize	int
aspectRatio	videoProfile.AspectRatio
frameRateDomain	rawVideo.FrameRate
audios	list of AudioTrack
subServices	list of SubService

40.3.8 MultiServiceProfileObj

MultiServiceProfile

40.3.9 SetMultiServiceProfiles.Request

struct

data **map** from **UUID** to **MultiServiceProfileObj**

40.3.10 SetMultiServiceProfiles.Response

empty **struct**

40.3.11 SubService

Subservice configuration parameters.
 @param index Subservice unique id. Values are not required to be contiguous.
 @param video Video parameters.
 @param audioSubscriptions List of AudioTrack uid's to include in this subservice. These must be

struct

index **int**

video	optional VideoTrack
audioSubscriptions	list of int

40.3.12 VideoTrack

Video Track configuration parameters.
@param profile Describes a video coding process.

struct

profile	videoProfile.VideoProfileStruct
---------	--

4.1 multiServiceProfileMode (1.0)

4.1.1 Type Reference

4.1.1.1 ProfileMode

enum

ABR

BROADCAST

42 nmosConfig (1.1)

42.1 Overview

```
# Changelog

## 1.1
- **Added**
  - Added new type `ConnectionType`
  - Added member 'connectionType' to NmosConfig
```

42.2 Type Reference

42.2.1 ConnectionType

Enum to control whether the NMOS traffic is in-band (through line card dataports) or out-of-band (through the MMI).

enum

```
IN_BAND
OUT_OF_BAND
```

42.2.2 NmosConfig

Configuration parameters for setting up NMOS.

struct

connectionType	ConnectionType Configuration to control whether traffic should be in-band or out-of-band
registry	optional nmosRegistryConfig.NmosRegistryConfigStruct Configuration parameters for connecting to the NMOS registry
labelTemplates	optional nmosLabelConfig.NmosLabelConfig Custom NMOS labels configuration. This feature is enabled when the optional is set

43 nmosLabelConfig (1.0)

43.1 Overview

Changelog

1.0

NMOS labelling

NMOS labelling API allows operator to define templates based on auto text options. This templates can be defined for node, device, source, flow, sender and receiver.

Version 1.0 allows six auto text options.

What are the auto text options?

- `${SLOT}` – The slot number of the card in use.
- `${DEV_ID}` – The Device number. **Not supported for Node templates.**
- `${ESS_GR}` – The label of the group where the essences belong to. **Not supported for Node templates.**
- `${ESS}` – The label of the Essence in question. **Not supported for Node and Device templates.**
- `${ESS_T}` – The Essence type: Video, Audio, Ancillary. This auto text option is substituted by what is defined in `EssenceTypeStrings` for each of the essences type. **Not supported for Node and Device templates.**
- `${ESS_T_INDEX}` – The number of the essence belonging to one of the essence type – video, audio, ancillary. **Not supported for Node and Device templates.**

How to use the NMOS Labelling.

- NMOS templates are set in the fields part of `ResourceLabelTemplates`. Unused templates should be set with an empty string.
- Each field in `ResourceLabelTemplates` can have as many auto text options as the operator wants, as long as they are supported for the respective template.
- `EssenceTypeStrings` strings will substitute the auto text option `${ESS_T}` with the defined string for the respective Essence type.

Example of strings to use for the fields in `ResourceLabelTemplates` and `EssenceTypeStrings`

- `nodeTemplate` – ECx210 Decoder ST2110 Sender Slot `${SLOT}`
- `deviceTemplate` – ECx210 Decoder Sender Slot `${SLOT}` ID `${DEV_ID}`
- `sourceTemplate` – ECx210 Slot `${SLOT}` ID `${DEV_ID}`, `${ESS_T}` Source: `${ESS}`, Group: `${ESS_GR}`, Essence Number: `${ESS_T_INDEX}`
- `flowTemplate` – ECx210 Slot `${SLOT}` ID `${DEV_ID}`, `${ESS_T}` Flow: `${ESS}`, Group: `${ESS_GR}`, Essence Number: `${ESS_T_INDEX}`
- `senderTemplate` – ECx210 Slot `${SLOT}` ID `${DEV_ID}`, `${ESS_T}` Sender: `${ESS}`, Group: `${ESS_GR}`, Essence Number: `${ESS_T_INDEX}`
- `receiverTemplate` – ECx210 Slot `${SLOT}` ID `${DEV_ID}`, `${ESS_T}` Receiver: `${ESS}`, Group: `${ESS_GR}`, Essence Number: `${ESS_T_INDEX}`
- `video` – Video
- `audio` – Audio
- `ancillary` – Anc

43.2 Type Reference

43.2.1 EssenceTypeStrings

Labels for each essence type to use when generating custom NMOS labels.

struct

video	string Video essence label
audio	string Audio essence label
ancillary	string Ancillary essence label

43.2.2 NmosLabelConfig

Custom NMOS labels configuration.

struct

resourceTemplates	ResourceLabelTemplates Templates for each NMOS resource
essenceStrings	EssenceTypeStrings Labels to use for each essence type

43.2.3 ResourceLabelTemplates

NMOS label templates for each NMOS resource.

struct

nodeTemplate	string Template string for the node label
deviceTemplate	string Template string for the device label
sourceTemplate	string Template string for the source label
flowTemplate	string Template string for the flow label
senderTemplate	string Template string for the sender label
receiverTemplate	string Template string for the receiver label

44 nmosRegistryConfig (1.0)

44.1 Type Reference

44.1.1 DnsConfig

Method to use for discovery of registry server

variant

mDns	DnsConfig.mDns Use mDNS to reach the registry server
dnsSd	DnsSdConfig Configuration parameters to use DNS-SD for discovery of registry service

44.1.2 DnsConfig.mDns

empty **struct**

44.1.3 DnsSdConfig

Configuration parameters for DNS-SD

struct

dnsServer	firewallTypes.IpAddress IPv4 address of DNS-SD server
search	string Search domain name

44.1.4 NmosRegistryConfigStruct

Configuration parameters for reaching the NMOS registry

struct

registryConfig	RegistryManualConfig Configuration parameters to use in case of failure in mDNS or DNS-SD (secondary/fallback method)
dnsConfig	DnsConfig Configuration parameters for mDNS or DNS-SD (primary method)

44.1.5 RegistryManualConfig

Registry information to be used in case registry server can not be reached by mDNS or DNS-SD.

struct

fallbackEnable	bool Enable use of given information as a fallback
	firewallTypes.IpAddress

registryIpAddr	IPv4 address of NMOS registry
registryPort	int Port number of NMOS registry
registryVersion	commonTypes.NmosVersionNumber Version of registry API

45 nmosRegistryStatus (1.0)

45.1 Type Reference

45.1.1 NmosRegistryStatus

NMOS registry status

struct

ipAddr

optional **firewallTypes.SocketAddress**

Currently used IP address and port number of the registry server. This value may have been obtained from DNS-SD, mDNS, or a manual IP entry.

registered

bool

True if there is an active connection to the registry server.

46 nmosStatusTypes (1.2)

46.1 Type Reference

46.1.1 NmosStatus

NmosStatus Status of the NMOS service

struct

nmosRegistryStatus	optional nmosRegistryStatus.NmosRegistryStatus Status of the NMOS registry connection. Empty if no registry is configured.
nodeApi	map from physicalIpPort.PortName to string Map from port to the endpoint for NMOS Node API. Ex: PortName.D1 -> "10.20.10.34:8001", PortName.D2_2 -> "10.20.11.34:8001"
activeOutOfBandPath	optional int The active out-of-band path. Optional: empty for in-band configuration, 0 if MMI slot 1, 1 if MMI slot 2.

47 physicallyPort (1.0)

47.1 Type Reference

47.1.1 PortName

enum

CTRL	
D1	
D2	
D3	
D4	
D1_1	
D1_2	
D1_3	
D1_4	
D2_1	
D2_2	
D2_3	
D2_4	
D3_1	
D3_2	
D3_3	
D3_4	
D4_1	
D4_2	
D4_3	
D4_4	

48 poisServerStatus (1.0)

48.1 Overview

```
# Changelog

## Version 1.0
- **Added**
  - `PoisServerStatus`
  - `GetPoisServerStatus`
```

48.2 Command Reference

48.2.1 GetPoisServerStatus

- message **GetPoisServerStatus.Request**
- message **GetPoisServerStatus.Response**

48.3 Type Reference

48.3.1 GetPoisServerStatus.Request

empty **struct**

48.3.2 GetPoisServerStatus.Response

struct

data **PoisServerStatus**

48.3.3 PoisServerStatus

Status of POIS server

@param connectionStatus Status of the connection to the server

struct

connectionStatus **esamStatus.PoisConnectionStatus**

49 poolConfig (1.1)

49.1 Overview

Changelog

1.1

- Changed
 - included esamConfig in PoolConfig struct

49.2 Command Reference

49.2.1 GetPoolConfig

- message **GetPoolConfig.Request**
- message **GetPoolConfig.Response**

49.2.2 SetPoolConfig

- message **SetPoolConfig.Request**
- message **SetPoolConfig.Response**

49.3 Type Reference

49.3.1 GetPoolConfig.Request

empty **struct**

49.3.2 GetPoolConfig.Response

struct

data **PoolConfigObj**

49.3.3 PoolConfig

struct

spareModules	optional int
esamConfig	optional esamConfig.EsamConfig

49.3.4 PoolConfigObj

PoolConfig

49.3.5 SetPoolConfig.Request

struct

data

PoolConfigObj

49.3.6 SetPoolConfig.Response

empty struct

50 poolCtrlAllocs (1.1)

50.1 Overview

Changelog:

1.1

- Added
 - Constant base UUID used to generate deterministic UUIDs for video alignment groups.
- Changed
 - generalization of the video alignment in order to work for both SDI and IP: VideoAlignGroupMaster, VideoAlignGroupMasterObj and FlowStatus should replace VideoAlignGroupMasters, Allocations and Distribution
 - SlotPort with member port changed to SlotFlow with member flow

50.2 Type Reference

50.2.1 SlotFlow

struct

slot	int
flow	int

50.2.2 VideoAlignGroupMaster

struct

groupIndex	int
master	UUID
flow	SlotFlow

50.2.3 VideoAlignGroupMasterObj

VideoAlignGroupMaster

51 psiStatus (1.6)

51.1 Overview

```
# Changelog

## 1.6
- **Changed**
  - `psiTypes` from 1.5 to 1.6

## 1.5
- **Changed**
  - `psiTypes` from 1.4 to 1.5
```

51.2 Type Reference

51.2.1 PsiStatus

struct

components	list of psiTypes.Component
name	optional string

52 psiTypes (1.6)

52.1 Overview

```
# Changelog

## 1.6
- **Changed**
  - Component.bitrate changed from int to bigint. Necessary for UHD

## 1.5
- **Changed**
  - `component` from 1.3 to 1.5
```

52.2 Type Reference

52.2.1 Component

struct

flowId	int
pid	Pid
lang	Lang
componentType	component.ComponentType
streamType	int
isPcr	bool
cas_ids	list of int
ccError	int
bitrate	bigint

52.2.2 Components

list of **Component**

52.2.3 Lang

string

52.2.4 Pid

int

53 rawAudio (1.2)

53.1 Overview

```
# Changelog
## 1.2
- **Added**
  - `SURROUND_514`, `SURROUND_714` and `SURROUND_916` to `ChannelMode`.
```

53.2 Type Reference

53.2.1 ChannelMode

enum

MONO	
DUALMONO	
STEREO	
SURROUND_31	
SURROUND_51	
SURROUND_71	
SURROUND_514	
SURROUND_714	
SURROUND_916	

53.2.2 Hack

struct

profile	int
---------	-----

54 rawVideo (1.5)

54.1 Type Reference

54.1.1 ChromaSampling

enum

CS_422

CS_420

54.1.2 FrameRate

enum

FPS_25

FPS_29_97

FPS_30

FPS_50

FPS_59_94

FPS_60

54.1.3 Hack

struct

profile

int

54.1.4 HorizontalResolution

enum

H_3840

H_2560

H_1920

H_1440

H_1280

H_1024

H_960

H_854

H_848

H_768

H_720

H_704

H_640

H_544

H_528

H_512

H_480

H_400

H_384

H_320

H_256

H_192

H_416

54.1.5 VerticalResolution

enum

V_2160

V_1440

V_1080

V_720

V_640

V_576

V_540

V_480

V_432

V_396

V_384

V_360

V_288

V_270

V_224

V_216

V_192

V_180

V_144

V_240

55 redundancyGroup (1.0)

55.1 Overview

```
# Changelog

## 1.0
- **Added**
  - added RedundancyMode
  - added RedundancyType
  - added RedundancyCard
  - added RedundancyGroupStruct
```

55.2 Command Reference

55.2.1 GetRedundancyGroups

- message **GetRedundancyGroups.Request**
- message **GetRedundancyGroups.Response**
- message **GetRedundancyGroups.Failure**

55.2.2 SetRedundancyGroups

- message **SetRedundancyGroups.Request**
- message **SetRedundancyGroups.Response**
- message **SetRedundancyGroups.Failure**

55.2.3 DeleteRedundancyGroups

- message **DeleteRedundancyGroups.Request**
- message **DeleteRedundancyGroups.Response**
- message **DeleteRedundancyGroups.Failure**

55.3 Type Reference

55.3.1 DeleteRedundancyGroups.Failure

empty **struct**

55.3.2 DeleteRedundancyGroups.Request

struct

ids **list** of **UUID**

55.3.3 DeleteRedundancyGroups.Response

empty **struct**

55.3.4 GetRedundancyGroups.Failure

empty **struct**

55.3.5 GetRedundancyGroups.Request

empty **struct**

55.3.6 GetRedundancyGroups.Response

struct

data **map** from **UUID** to **RedundancyGroup**

55.3.7 RedundancyCard

N+M redundancy configuration for a single card

struct

slot	int The slot the card resides in
redundancyType	RedundancyType Specifies whether the card is a main or backup card
disabled	bool True if the card has been disabled

55.3.8 RedundancyGroup

RedundancyGroupStruct

55.3.9 RedundancyGroupStruct

N+M redundancy group configuration

struct

label	string A label for this group
router	optional UUID The SDI router connected to the cards in this group
mode	RedundancyMode The redundancy mode used for this group
cards	list of RedundancyCard List of cards that are part of this group

55.3.10 RedundancyMode

The supported N+M redundancy modes

enum

FLOATING	
REVERTING	

55.3.11 RedundancyType

The roles a card can have in a redundancy group

enum

MAIN	
BACKUP	

55.3.12 SetRedundancyGroups.Failure

empty struct

55.3.13 SetRedundancyGroups.Request

struct

data map from UUID to RedundancyGroup

55.3.14 SetRedundancyGroups.Response

empty struct

56 redundancyGroupStatus (1.2)

56.1 Overview

```
# Changelog

## 1.2
- **Added**
  - added redundancyStateToString

## 1.1
- **Added**
  - added ProtoRedundancyGroupStatus
```

56.2 Command Reference

56.2.1 GetRedundancyGroupStatus

- message **GetRedundancyGroupStatus.Request**
- message **GetRedundancyGroupStatus.Response**

56.3 Type Reference

56.3.1 GetRedundancyGroupStatus.Request

empty **struct**

56.3.2 GetRedundancyGroupStatus.Response

struct

data **map** from **UUID** to **RedundancyGroupStatus**

56.3.3 RedundancyCardStatus

N+M redundancy status for a single card

struct

state	RedundancyState The current state of the card
activeConfig	optional Slot The active configuration of the card. A slot means that the services on that slot was moved to this card.

56.3.4 RedundancyGroupStatus

N+M redundancy status for a redundancy group

struct

map from **Slot** to **RedundancyCardStatus**

status

Redundancy status for each card in this group

56.3.5 RedundancyState

The states a single card can be in.

enum

UP

DOWN

DISABLED

56.3.6 Slot

int

57 scte35Config (1.0)

57.1 Type Reference

57.1.1 Scte35Config

Config for a Scte35 configuration.

struct

idrOnScte35	bool Enables IDR insertion on SCTE 35 splice points.
esamConfig	optional esamConfig.EsamServiceConfig ESAM configuration. A set optional indicates that ESAM is enabled.

58 scte35LogApi (1.0)

58.1 Command Reference

58.1.1 GetScte35History

- message **GetScte35History.Request**
- message **GetScte35History.Response**

58.2 Type Reference

58.2.1 GetScte35History.Request

struct

serviceId	UUID
slot	int

58.2.2 GetScte35History.Response

struct

scte35History	string
---------------	--------

59 sdiConnection (1.0)

59.1 Overview

New in v1.0:

- These types were factored out from coderService

59.2 Type Reference

59.2.1 SdiConnection

struct

sdiType	sdiType.SdiType
connectors	SdiConnectors

59.2.2 SdiConnectors

enum

SDI_1A	
SDI_1B	
SDI_1C	
SDI_1D	
SDI_2A	
SDI_2B	
SDI_2C	
SDI_2D	
SDI_1ABCD	
SDI_2ABCD	
SDI_1AC	
SDI_1BD	
SDI_2AC	
SDI_2BD	
UNDEFINED	

60 sdiRouterConfig (1.1)

60.1 Overview

```
# Changelog
## 1.1
- **Added**
  - added brand SWP08
  - added port to SdiRouterConfig

## 1.0
- **Added**
  - added SdiRouterStruct
  - added SdiRouterBrand
  - added CardConnector
```

60.2 Command Reference

60.2.1 GetSdiRouters

- message **GetSdiRouters.Request**
- message **GetSdiRouters.Response**
- message **GetSdiRouters.Failure**

60.2.2 SetSdiRouters

- message **SetSdiRouters.Request**
- message **SetSdiRouters.Response**
- message **SetSdiRouters.Failure**

60.2.3 DeleteSdiRouters

- message **DeleteSdiRouters.Request**
- message **DeleteSdiRouters.Response**
- message **DeleteSdiRouters.Failure**

60.3 Type Reference

60.3.1 CardConnector

Holds slot and connector which is enough to specify a distinct input port in a chassis.

struct

slot	int
connector	sdiConnection.SdiConnectors

60.3.2 DeleteSdiRouters.Failure

empty **struct**

60.3.3 DeleteSdiRouters.Request

struct

ids **list of UUID**

60.3.4 DeleteSdiRouters.Response

empty **struct**

60.3.5 GetSdiRouters.Failure

empty **struct**

60.3.6 GetSdiRouters.Request

empty **struct**

60.3.7 GetSdiRouters.Response

struct

data **map from UUID to SdiRouter**

60.3.8 Label

string

60.3.9 Port

int

60.3.10 SdiRouter

SdiRouterStruct

60.3.11 SdiRouterBrand

The different SDI router brands we support

enum

BLACKMAGIC

NEVION

SWP08

60.3.12 SdiRouterStruct

SDI router configuration

struct

label	string A label for the SDI router
ip	firewallTypes.IpAddress The IP address of the router
port	optional int The port of the router
brand	SdiRouterBrand The router brand
numInputs	int The number of SDI inputs the router has
numOutputs	int The number of SDI outputs the router has
inputs	map from Port to Label Label for each input port
outputs	map from Port to Label Label for each output port
outputMapping	map from Port to CardConnector Maps router output port to coder slot and connector

60.3.13 SetSdiRouters.Failureempty **struct****60.3.14 SetSdiRouters.Request****struct**data **map** from **UUID** to **SdiRouter****60.3.15 SetSdiRouters.Response**empty **struct**

61 sdiRouterConnection (1.0)

61.1 Overview

```
# Changelog

## 1.0
- **Added**
  - added SdiRouterConnectionStruct
```

61.2 Command Reference

61.2.1 SetSdiRouterConnections

- message **SetSdiRouterConnections.Request**
- message **SetSdiRouterConnections.Response**
- message **SetSdiRouterConnections.Failure**

61.3 Type Reference

61.3.1 InputPort

int

61.3.2 OutputPort

int

61.3.3 SdiRouterConnectionStruct

Keeps track of the internal routing in a SDI router

struct

routerId	UUID The ID of the SDI router
routing	map from OutputPort to InputPort Map of which input is available on each output

61.3.4 SetSdiRouterConnections.Failure

empty **struct**

61.3.5 SetSdiRouterConnections.Request

struct

data	map from UUID to SdiRouterConnectionStruct
------	---

61.3.6 SetSdiRouterConnections.Response

empty **struct**

62 sdiRouterStatus (1.0)

62.1 Overview

```
# Changelog

## 1.0
- **Added**
  - added SdiRouterConfigStatus
  - added GetSdiRouterConfigStatuses
```

62.2 Command Reference

62.2.1 GetSdiRouterStatuses

- message **GetSdiRouterStatuses.Request**
- message **GetSdiRouterStatuses.Response**

62.3 Type Reference

62.3.1 GetSdiRouterStatuses.Request

empty **struct**

62.3.2 GetSdiRouterStatuses.Response

struct

data **map** from **UUID** to **SdiRouterStatus**

62.3.3 InputPort

int

62.3.4 Label

string

62.3.5 OutputPort

int

62.3.6 SdiRouterStatus

Status for a connected SDI router

@param routing The currently configured routing for each output on the router
@param inputLabels The labels for the inputs on the switch
@param outputLabels The labels for the outputs on the switch

struct

routing	map from OutputPort to InputPort
inputLabels	map from InputPort to Label
outputLabels	map from OutputPort to Label
link	bool

63 sdiStatus (1.11)

63.1 Overview

```
# Changelog
## 1.11
- **Changed**
  - `codedAudio` from 1.0 to 1.1

## 1.10
- **Added**
  - VPID status to SDI status

## 1.9
- **Added**
  - SdiAudioStatus analysedMaxLevel and analysedCodec.

## 1.8
- **Added**
  - SDI_2160P25, SDI_2160P30, SDI_2160P30_1001 to the SdiVideoModes enum
```

63.2 Type Reference

63.2.1 AesChannelMode

enum

MONO
STEREO
MULTI
N_A
UNKNOWN

63.2.2 AesData

struct

mode	AesMode
channelMode	AesChannelMode
crcOk	bool

63.2.3 AesMode

enum

PCM
NON_PCM

63.2.4 InstanceId

int

63.2.5 QuadMode

enum

SDI_4X3G_SQUARE	
SDI_4X3G_INTERL	

63.2.6 QuadStatus

struct

lock	bool
lockLoss	int
skew	float

63.2.7 SdiAudioSampleRate

enum

SYNC_48_kHz	
ASYNC_48_kHz	
SYNC_44_1_kHz	
ASYNC_44_1_kHz	
SYNC_32_kHz	
ASYNC_32_kHz	
SYNC_FREE_RUN	
ASYNC_FREE_RUN	

63.2.8 SdiAudioStatus

SDI input audio status on an encoder.

struct

aesChannel	int Embedded audio channels.
active	bool Indicates if a channel is active meaning that there exists a valid signal on the input.
lossCnt	int Number of lost packets during the last second.
sampleRate	SdiAudioSampleRate Audio sample rate.
sampleCnt	int Actual number of samples received during the last second.
errors	bool True if any errors during the last second, false otherwise.
aesData	optional AesData Embedded audio channel data.
analysedMaxLevel	optional float Analysed input max level.

analysedCodec

optional codedAudio.AudioCodec
Analysed input codec.

63.2.9 SdiStatus

SDI input status on an encoder, and SDI output status on a decoder.

struct

id	InstanceId Used with quad SDI input. IDs 0-3 correspond to signals on 1A-D and 4-7 correspond to 2A-D.
videoLockInput	bool True if the encoder has locked onto a signal, false otherwise.
videoMode	SdiVideoModes Input video mode on encoder, and output video mode on the decoder.
edhErrorCount	int Number of EDH errors on the input signal.
audio	list of SdiAudioStatus List of audio statuses for each audio channel in the SDI signal.
quad	optional QuadStatus Quad status for this SDI instance. Only available with quad input.
quadMode	optional QuadMode Which mode was used to generate the quad signal. Only available with quad input.
transceiverStatus	optional transceiverStatus.TransceiverStatus Status from the transceiver. Only available on encoder.
vpidStatus	optional vpidInfo.ColorInfo VPID color info going out of the decoder, and coming in on the encoder.

63.2.10 SdiVideoModes

enum

SDI_OFF
 SDI_480I
 SDI_480P
 SDI_576I
 SDI_576P
 SDI_720P24
 SDI_720P24_1001
 SDI_720P25
 SDI_720P30
 SDI_720P30_1001
 SDI_720P50
 SDI_720P60
 SDI_720P60_1001

SDI_1080P24

SDI_1080P24_1001

SDI_1080P25

SDI_1080P30

SDI_1080P30_1001

SDI_1080I50

SDI_1080I60

SDI_1080I60_1001

SDI_1080P50

SDI_1080P60

SDI_1080P60_1001

SDI_2160P50

SDI_2160P60

SDI_2160P60_1001

SDI_2160P25

SDI_2160P30

SDI_2160P30_1001

ANY

64 sdiType (1.0)

64.1 Type Reference

64.1.1 sdiType

enum

SDI_3G

SDI_12G

SDI_4X3G_QUANT

SDI_4X3G_SQUARE

SDI_4X3G_INTERL

65 sdpUpload (1.1)

65.1 Overview

This file defines the external RPC interface for uploading SDP files in order to change an IP Pr

Changelog

1.1

- ****Changed****
- `profileId` added to `SdpData` struct for use with Profile Redundancy

65.2 Command Reference

65.2.1 UploadSdpData

- message **UploadSdpData.Request**
- message **UploadSdpData.Response**

65.3 Type Reference

65.3.1 SdpData

- SDP data for storing the SDP file uploaded through GUI.

•

struct

serviceId

UUID

The uuid of the service for which the SDP file is uploaded.

•

profileId

optional UUID

The uuid of the profile for which the SDP file is uploaded. If profileId is empty, the uploaded SDP should belong to the first profile (index 0).

•

slot

int

The slot on which this SDP data will be used.

•

sdpInfo

map from UUID to string

The SDP file stored with a unique uuid.

65.3.2 UploadSdpData.Request

SdpData

65.3.3 UploadSdpData.Response

empty **struct**

66 serviceStatus (2.43)

66.1 Overview

Changelog:

2.43

- **Changed**
 - lockStatus from 1.3 to 1.4
 - audioStatus from 2.14 to 2.15
 - st2110Status from 1.4 to 1.5
 - sdiStatus from 1.10 to 1.11

2.42

- **Changed**
 - videoStatus from 2.14 to 2.15
 - vancStatus from 1.12 to 1.13

2.41

- **Changed**
 - videoStatus from 2.13 to 2.14

2.40

- **Changed**
 - videoStatus from 2.12 to 2.13
 - sdiStatus from 1.9 to 1.10
 - vancStatus from 1.11 to 1.12

2.39

- **Changed**
 - vancStatus from 1.10 to 1.11
 - audioStatus from 2.13 to 2.14
 - st2110Status from 1.3 to 1.4

2.38

- **Changed**
 - vancStatus from 1.9 to 1.10

2.37

- **Changed**
 - videoStatus from 2.11 to 2.12

2.36

- **Changed**
 - vancStatus from 1.8 to 1.9

2.35

- **Changed**
 - st2110Status from 1.2 to 1.3
 - audioStatus from 2.12 to 2.13
 - vancStatus from 1.7 to 1.8

2.34

- **Changed**
 - audioStatus from 2.11 to 2.12
 - sdiStatus from 1.8 to 1.9

2.33

- **Changed**
 - vancStatus from 1.6 to 1.7

2.32

- **Changed**
 - vancStatus from 1.5 to 1.6

2.31

- **Changed**
 - lockStatus from 1.2 to 1.3

2.30

- **Changed**
 - videoStatus from 2.10 to 2.11
 - psiStatus from 1.5 to 1.6
 - inputRedundancyStatus from 1.2 to 1.3

2.29

- **Added**
 - New struct St2110ProfileStatus added, replacing st2110 list in ServiceStatus

2.28

- **Changed**
 - sdiStatus from 1.7 to 1.8

2.27

- **Changed**
 - audioStatus from 2.10 to 2.11
 - psiStatus from 1.4 to 1.5
 - inputRedundancyStatus from 1.1 to 1.2

2.26

- **Changed**

– ipStatus from 1.6 to 1.7

66.2 Command Reference

66.2.1 GetServiceStatus

- message **GetServiceStatus.Request**
- message **GetServiceStatus.Response**

66.3 Type Reference

66.3.1 GetServiceStatus.Request

ServiceStatusRequest

66.3.2 GetServiceStatus.Response

ServiceStatusResponse

66.3.3 ServiceStatus

Status for a specific service

struct

video	optional videoStatus.VideoStatus Status for resulting encoded or decoded video.
audio	optional audioStatus.AudioStatus Audio status
sdi	map from int to sdiStatus.SdiStatus SDI input status per channel (also used for IP-2022-Encoder)
st2110	optional st2110ProfileStatus ST-2110 specific status; list of essences and uuid of the active ip profile
lock	optional lockStatus.LockStatus Lock status (genlock etc.)
vanc	optional vancStatus.VancStatus Status for the supported VANC protocols (teletext etc.)
ip	map from int to ipStatus.IpStatus IP status
components	optional psiStatus.PsiStatus Status for component passthrough (transcoder)
latency	optional int Total latency of the service in ms (encoder only)
tsInputRedundancy	optional inputRedundancyStatus.TsInputRedundancyStatus

66.3.4 ServiceStatusQuery

struct

id	UUID
slot	int

66.3.5 ServiceStatusRequest

struct

query	list of ServiceStatusQuery
-------	----------------------------

66.3.6 ServiceStatusResponse

struct

data	map from UUID to ServiceStatus
------	--------------------------------

66.3.7 St2110ProfileStatus

struct

st2110Essences	list of st2110Status.St2110Status
profileId	UUID

67 sfpConstants (1.4)

67.1 Overview

Changelog

1.4

- **Changed**
- Changed "SFP or SFP+" to "SFP/SFP+/SFP28" The corresponding identifier refers to the given 3 SFP types as per SNIA SFF-8024 specification.

67.2 Type Reference

67.2.1 Identifier

Physical Device Identifier Values. (SFF 8472 Rev12.3 spec Table 5-1)

enum

UNKNOWN	Unknown or unspecified
GBIC	GBIC
SOLDERED	Module soldered to motherboard (ex. SFF)
SFP_SFPP	SFP, SFP+ or SFP28
QSFP	QSFP+
QSFP28	QSFP28
NOT_USED	Not used by SFF 8472 Rev12.2.1. These values are maintained in the Transceiver Management section of SFF-8024.
VENDOR	Vendor specific

67.2.2 Transceiver

Transceiver Compliance Codes. (SFF 8472 Rev12.3 spec Table 5-3)

enum

ER_10G	10G Base-ER
LRM_10G	10G Base-LRM
LR_10G	10G Base-LR
SR_10G	10G Base-SR
ACTIVE_CABLE_40G	40G Active cable
LR4_40G	40G Base-LR4
SR4_40G	40G Base-SR4
CR4_40G	40G Base-CR4
PX	BASE-PX
BX10	BASE-BX10
FX_100	100BASE-FX
LX_100	100BASE-LX/LX10
T_1000	1000BASE-T
CX_1000	1000BASE-CX
LX_1000	1000BASE-LX
SX_1000	1000BASE-SX

ELECTRICAL_INTER_ENCLOSURE	Electrical inter enclosure
LONGWAVE_LASER_LL	Longwave laser LL
MEDIUM	Medium distance
LONG_DISTANCE	Long distance
INTERMEDIATE_DISTANCE	Intermediate distance
SHORT_DISTANCE	Short distance
VERY_LONG_DISTANCE	Very long distance
ACTIVE_CABLE	Active Cable
PASSIVE_CABLE	Passive Cable
LONGWAVE_LASER_LC	Longwave laser LC
SHORTWAVE_LASER_w_OFC	Shortwave laser w OFC
SHORTWAVE_LASER_wo_OFC	Shortwave laser w/o OFC
ELECTRICAL_INTRA_ENCLOSURE	Electrical intra enclosure
SINGLE_MODE	Singel mode
MULTIMODE_50um	Multimode 50um
MULTIMODE_50m	Multimode 50m
MULTIMODE_625m	Multimode 62.6m
VIDEO_COAX	Video coax
MINIATURE_COAX	Miniature coax
SHIELDED_TWISTED_PAIR	Shielded twisted pair
TWIN_AXIAL_PAIR	Twin axial pair
M_BYTES_SEC_100	100M bytes per sec
M_BYTES_SEC_200	200M bytes per sec
M_BYTES_SEC_400	400M bytes per sec
M_BYTES_SEC_1600	1600M bytes per sec
M_BYTES_SEC_800	800M bytes per sec
M_BYTES_SEC_1200	1200M bytes per sec
AOC_100G_AOC_25GAUI_C2M_BER_5_	Extended Specification Compliance. (SFF-8024 Rev 4.8 spec Table 4-4)
AOC_100G_AOC_25GAUI_C2M_BER_5_	100G AOC (Active Optical Cable) or 25GAUI C2M AOC. Providing a worst BER of $5 * 10^{-5}$
SR4_100G_BASE_SR_25G_BASE	100GBASE-SR4 or 25GBASE-SR
LR4_100G_BASE_LR_25G_BASE	100GBASE-LR4 or 25GBASE-LR
ER4_100G_BASE_ER_25G_BASE	100GBASE-ER4 or 25GBASE-ER
SR10_100G_BASE	100GBASE-SR10
CWDM4_100G	100G CWDM4
PSM4_PARALLEL_SMF_100G	100G PSM4 Parallel SMF
ACC_100G_ACC_25GAUI_C2M_BER_5_	100G ACC (Active Copper Cable) or 25GAUI C2M ACC. Providing a worst BER of $5 * 10^{-5}$
CR4_100G_BASE_25G_BASE_CR_CA_2	100GBASE-CR4 or 25GBASE-CR CA-L
CR_25G_BASE_CA_25G_S_50G_BASE_	25GBASE-CR CA-S
CR_25G_BASE_CA_25G_N_50G_BASE_	25GBASE-CR CA-N
ER4_40G_BASE	40GBASE-ER4
SR_4_X_10G_BASE	4 x 10GBASE-SR
PSM4_PARALLEL_SMF_40G	40G PSM4 Parallel SMF
G959_1_PROFILE_P1I1_2D1	G959.1 profile P1I1-2D1 (10709 MBd, 2km, 1310nm SM)
G959_1_PROFILE_P1S1_2D2	G959.1 profile P1S1-2D2 (10709 MBd, 40km, 1550nm SM)
G959_1_PROFILE_P1L1_2D2	G959.1 profile P1L1-2D2 (10709 MBd, 80km, 1550nm SM)
T_10G_BASE	10GBASE-T with SFI electrical interface

CLR4_100G	100G CLR4
AOC_100G_AOC_25GAUI_C2M_BER_10	100G AOC or 25GAUI C2M AOC. Providing a worst BER of 10^{-12} or below
ACC_100G_ACC_25GAUI_C2M_BER_10	100G ACC or 25GAUI C2M ACC. Providing a worst BER of 10^{-12} or below
DWDM2_100GE	100GE-DWDM2 (DWDM transceiver using 2 wavelengths on a 1550nm DWDM grid with a reach up to 80km)

68 sfpStatus (1.5)

68.1 Overview

Changelog

1.5

- Changed
 - sfpConstants from 1.3 to 1.4

68.2 Type Reference

68.2.1 SfpDiagnostics

Real Time Diagnostic for SFP/SFP+/QSFP28/QSFP+, with A2h/A0h memory space address in square brackets. (SFF 8472 Rev12.3 spec Table 9-11) (SFF 8636 Rev2.10 spec Table 6-1)

struct

temp	float Internally measured module temperature (°C). [96-97], [22-23]
vcc	float Internally measured supply voltage in transceiver (V). [98-99], [26-27]
txPwr	list of float Measured TX output power (mW). [102-103], [50-57]
rxPwr	list of float Measured RX input power (mW). [104-105], [34-41]

68.2.2 SfpStatus

SFP Status Data Fields, with A0h memory space address in square brackets. (SFF 8472 Rev12.3 spec Table 4-1) QSFP A0h memory space address is specified next to the SFP A0h. (SFF 8636 Rev 2.10 spec Table 6-14)

struct

identifier	sfpConstants.Identifier Type of transceiver (SFF 8024 Rev4.8a spec Table 4-1) [0],[128]
transceiver	list of sfpConstants.Transceiver Code for electronic or optical compatibility [3-10], [131-138]
brNominal	float Nominal signalling rate, units of 100 MBd. [12], [140/222]
vendorName	string SFP vendor name. [20-35], [148-163]
vendorOUI	string SFP vendor IEEE company ID. [37-39], [165-167]
vendorPN	string Part number provided by SFP vendor. [40-55], [168-183]
	string

vendorRev	Revision level for part number provided by vendor. [56-59], [184-185]
wavelength	string Laser wavelength. [60-61], [186-187]
vendorSN	string Serial number provided by SFP vendor. [68-83], [196-211]
dateCode	string Vendor's manufacturing date code. [84-91], [212-219]
diagnostics	optional SfpDiagnostics Optional Diagnostics Monitor Data.

69 slateInsertion (1.0)

69.1 Overview

```
# Changelog

## 1.0
- **Added**
  - First version of module
```

69.2 Type Reference

69.2.1 SlateInsertion

Configure slate insertion for a CoderService.

struct

trigger	SlateInsertionTrigger What events will trigger slate insertion
slateId	string A unique identifier for the slate

69.2.2 SlateInsertionTrigger

enum

DPI
FORCED

70 slotInterface (1.6)

70.1 Overview

- This file defines the external RPC interface to forward
- GetSlotInterface request to the correct card.

Changelog:

1.6

- Changed
 - ipInterface from 1.5 to 1.6

1.5

- Changed
 - ipInterface from 1.4 to 1.5

70.2 Command Reference

70.2.1 GetSlotInterfaces

- message `GetSlotInterfaces.Request`
- message `GetSlotInterfaces.Response`
- message `GetSlotInterfaces.Failure`

70.3 Type Reference

70.3.1 GetSlotInterfaces.Failure

empty `struct`

70.3.2 GetSlotInterfaces.Request

`IpInterfaceRequest`

70.3.3 GetSlotInterfaces.Response

`struct`

data `map` from `UUID` to `ipInterface.IpInterface`

70.3.4 IpInterfaceRequest

`struct`

slot `int`

71 st2110Status (1.5)

71.1 Overview

```
# Changelog
## 1.4
- **Changed**
  - `audioProfile` from 2.5 to 2.6

## 1.4
- **Changed**
  - `audioProfile` from 2.4 to 2.5

## 1.3
- **Changed**
  - `st2110Types` from 1.0 to 1.2.
  - `audioProfile` from 2.3 to 2.4
- **Removed**
  - `St2110AudioPacketTime` moved to `st2110Types` 1.2.
```

71.2 Type Reference

71.2.1 St2110AncStatus

struct

ipStreamUuid	UUID
standard	st2110Types.St2110Standard
sdpStatus	list of string

71.2.2 St2110AudioStatus

struct

ipStreamUuid	UUID
standard	st2110Types.St2110Standard
packetTime	st2110Types.St2110AudioPacketTime
totalChannels	int
sampleRate	optional audioProfile.SampleRate
sdpStatus	list of string

71.2.3 St2110Status

variant

videoStatus	St2110VideoStatus
audioStatus	St2110AudioStatus
ancStatus	St2110AncStatus

71.2.4 St2110VideoStatus

struct

ipStreamUuid	UUID
standard	st2110Types.St2110Standard
sdpStatus	list of string

72 st2110Types (1.2)

72.1 Overview

```
# Changelog

## 1.2
- **Added**
- `St2110AudioPacketTime`
```

72.2 Type Reference

72.2.1 St2110AudioPacketTime

Valid audio packet time according to SMPTE ST2110.

enum

N_A	Not applicable
US83_3	83.3 us
US125	125 us
MS1	1ms

72.2.2 St2110Standard

The SMPTE ST2110 standards that describe video, audio or ancillary content.

enum

ST2110_20	ST2110 uncompressed video standard
ST2110_22	ST2110 compressed video standard
ST2110_30	ST2110 audio standard
ST2110_31	ST2110 AES3 formatted audio
ST2110_40	ST2110 ancillary standard

73 streamSource (1.0)

73.1 Type Reference

73.1.1 StreamSource

StreamSource types of stream sources.
@param MAIN the main stream source.
@param BACKUP the backup stream source.

enum

MAIN
BACKUP

74 subtitling (1.2)

74.1 Overview

```
# Changelog

## 1.2
- **Added**
  - New enum `HorizontalAlignment`.

## 1.1
- **Changed**
  - `component` from 1.3 to 1.5
```

74.2 Type Reference

74.2.1 HorizontalAlignment

enum

FOLLOW_INPUT	
LEFT	
CENTER	
RIGHT	

74.2.2 SubtitleComponentPattern

Transport stream subtitle source.

struct

language	string Three letter language code.
component	optional component.SubtitleFormat Subtitle format.
pid	optional int Subtitle PID.

75 testGenComponent (1.1)

75.1 Overview

New in v1.1:
- Use version 1.1 of testGeneratorProfile that includes Freeze Frame
New in v1.0:
- These types were factored out from coderService

75.2 Type Reference

75.2.1 TestGenComponent

Configuration for a test pattern generator profile.
@param id UUID of the test pattern generator profile containing the configuration.
@param cpy A copy of the content of the test pattern generator profile.
@param enable Parameter to force test generation to override any input

struct

profile	TestGenComponent.profile
enable	bool

75.2.2 TestGenComponent.profile

struct

id	UUID
cpy	optional testGeneratorProfile.TestGeneratorProfile

76 testGenerator (1.1)

76.1 Overview

```
# Changelog

## 1.1
- **Changed**
  - Added Freeze Frame as a signal loss option
```

76.2 Type Reference

76.2.1 BitmapBackgroundProperty

Specifies the color property of the background in a bitmap.

enum

BLACK	Make the bitmap background solid black.
TRANSPARENT	Make the bitmap background transparent, such that the bitmap in essence becomes an overlay.

76.2.2 Color

A color defined in terms of RGB.

struct

red	int Amount of red. Value 0 to 255.
green	int Amount of green. Value 0 to 255.
blue	int Amount of blue. Value 0 to 255.

76.2.3 SignalLoss

The types of output that can be produced in the event of signal loss.

enum

COLOR_BAR	Displays color bar image.
BLACK_FRAME	Displays black frame.
FREEZE_FRAME	Displays last frame available.
SYNC_LOSS	Displays nothing.
TEST_GENERATOR	

76.2.4 TextPosition

Specify the vertical placement for the text.

enum

TOP	Place it at the top of the image.
MIDDLE	Place it in the middle of the image.
BOTTOM	Place it at the bottom of the image.

77 testGeneratorProfile (1.1)

77.1 Overview

New in v1.1:
- update to 1.1 version of testGenerator (includes Freeze Frame)

77.2 Command Reference

77.2.1 GetTestGeneratorProfiles

- message **GetTestGeneratorProfiles.Request**
- message **GetTestGeneratorProfiles.Response**
- message **GetTestGeneratorProfiles.Failure**

77.2.2 SetTestGeneratorProfiles

- message **SetTestGeneratorProfiles.Request**
- message **SetTestGeneratorProfiles.Response**
- message **SetTestGeneratorProfiles.Failure**

77.2.3 DeleteTestGeneratorProfiles

- message **DeleteTestGeneratorProfiles.Request**
- message **DeleteTestGeneratorProfiles.Response**
- message **DeleteTestGeneratorProfiles.Failure**

77.3 Type Reference

77.3.1 DeleteTestGeneratorProfiles.Failure

empty **struct**

77.3.2 DeleteTestGeneratorProfiles.Request

struct

ids **list of UUID**

77.3.3 DeleteTestGeneratorProfiles.Response

empty **struct**

77.3.4 GetTestGeneratorProfiles.Failure

empty **struct**

77.3.5 GetTestGeneratorProfiles.Request

empty **struct**

77.3.6 GetTestGeneratorProfiles.Response

struct

data	map from UUID to TestGeneratorProfile
info	GetTestGeneratorProfiles.Response.info

77.3.7 GetTestGeneratorProfiles.Response.info

struct

totalEntries	int
--------------	-----

77.3.8 MovingBox

Parameters for specifying properties of test generation moving box
 @param enable Enable moving box.
 @param color Enum describing color of moving box.

struct

enable	bool
color	testGenerator.Color

77.3.9 SetTestGeneratorProfiles.Failure

empty struct

77.3.10 SetTestGeneratorProfiles.Request

struct

data	map from UUID to TestGeneratorProfile
------	---------------------------------------

77.3.11 SetTestGeneratorProfiles.Response

empty struct

77.3.12 TestGenVariant

Variant for the different types of test generation.

variant

lipSync	TestGenVariant.lipSync variant specifies that the profile activates lipSync.
testGenerator	TestGenerator variant contains test generation parameters.

77.3.13 TestGenVariant.lipSync

empty **struct**

77.3.14 TestGenerator

Properties of test generation

struct

testPattern	testGenerator.SignalLoss Enum describing type of pattern to be generated.
testTone	bool Audio tone generation for channels.
text	Text Parameters regarding adding text to the test generation.
box	MovingBox Parameters regarding unsng a moving box for the test generation.

77.3.15 TestGeneratorProfile

TestGeneratorProfileStruct

77.3.16 TestGeneratorProfileStruct

top level profile structure

struct

label	string Name of the profile
enable	bool Parameter to force test generation to override any input.
testGenVariant	TestGenVariant either lipSync or test generator.

77.3.17 Text

Parameters for specifying properties of test generation text
 @param enable Enable text.
 @param position Specifies screen location for the text.
 @param backgroundType Specifies the color property of the background.

struct

enable	bool
position	testGenerator.TextPosition
backgroundType	testGenerator.BitmapBackgroundProperty

78 timeMode (1.0)

78.1 Type Reference

78.1.1 TimeMode

Specifies the source of timing information.

enum

NTP

PTP_IF_LOCKED

PTP_FORCED

79 timecodeConfig (1.0)

79.1 Type Reference

79.1.1 TimecodeSource

The clock source to be used by the encoder.

enum

OFF

SMPTE_12M

PTP

NTP

80 timecodeStatus (1.0)

80.1 Type Reference

80.1.1 Timecode

- Timecode information from SMPTE 12M, PTP or NTP source.

struct

timecode	UnixTime Current time in seconds in unix format
isPresent	bool Availability of timecode information

80.1.2 UnixTime

bigint

81 transceiverStatus (1.1)

81.1 Type Reference

81.1.1 CableLengthIndicator

Information about the cable length indicator measurement. The measurement is given in terms of the Belden 1694A cable.

struct

length	int An indication of the cable length carrying the SDI input signal [m].
lengthPrecision	int The precision of the cable length measurement [m]. The actual length of the cable is likely to be in the interval [length - precision, length + precision].
maxLength	int The maximum supported cable length [m].
threshold	int The threshold below which measurements of cable length are very imprecise [m].

81.1.2 DataRate

The various input data rates the transceiver can lock onto.

enum

DR_11_88_GBPS
DR_5_94_GBPS
DR_2_97_GBPS
DR_1_485_GBPS
DR_270_MBPS
NO_LOCK

81.1.3 SignalProperties

struct

dataRate	DataRate An indication of the data rate of the SDI signal that the transceiver has locked.
cableLength	optional CableLengthIndicator An indication of the cable length carrying the SDI input signal [m]. N/A if dataRate is NO_LOCK.
horizontalEyeOpening	optional float Horizontal eye opening of the equalized SDI signal [UI]. Not available if dataRate is NO_LOCK. Not available for SD.

81.1.4 TransceiverStatus

Status about the quality of an SDI signal. These measurements are taken by the transceiver which is connected to the physical connector carrying the SDI signal.

struct

carrierDetect	bool True if the transceiver has detected that there is an input signal from an SDI cable.
signalProperties	optional SignalProperties Properties of the input signal that are available if a carrier is detected.

82 tsDestination (1.0)

82.1 Overview

New in v1.0:

- These types were factored out from coderService
- Was TsOut before. Got renamed to TsDestination

82.2 Type Reference

82.2.1 TsDestination

struct

serviceName	string
serviceId	int
serviceProvider	string
pcrInterval	float
pcrPid	int
pmtPid	int

83 tsSource (1.5)

83.1 Overview

```
# Changelog

## 1.5
- **Changed**
  - `subtitling` from 1.1 to 1.2

## 1.4
- **Added**
  - Amend `switchbackDelay` documentation in `InputRedundancy`; milliseconds, not seconds.
  - `switchDelay` to `InputRedundancy`

## 1.3
- **Added**
  - `FLOATING` to `SwitchingMode`.

## 1.2
- **Changed**
  - `subtitling` from 1.0 to 1.1
```

83.2 Type Reference

83.2.1 HotStandby

HotStandby configuration parameters.
 @param sourceProperty specifies whether the stream sources are identical or not.

struct

sourceProperty **SourceProperty**

83.2.2 IdenticalSources

IdenticalSources configuration parameters.
 @param maxOffset The maximum tolerated delay between identical sources in ms.

struct

maxOffset **int**

83.2.3 InputRedundancy

InputRedundancy configuration parameters.
 @param mode the input redundancy switching mode.
 @param switchbackDelay a delay in milliseconds to wait before reverting to main. N/A for switching
 @param switchDelay A delay in milliseconds to wait before switching to non-active source. N/A for
 @param manualOverride a chosen preferred source will be forced even if it is not present.
 @param hotStandby the hot standby configuration parameters.

struct

mode **SwitchingMode**

switchbackDelay	int
switchDelay	int
manualOverride	optional streamSource.StreamSource
hotStandby	HotStandby

83.2.4 SourceProperty

SourceProperty switching mode.
 @param nonIdenticalSources the stream sources are not identical.
 @param identicalSources identical stream sources configuration parameters.

variant

nonIdenticalSources	SourceProperty.nonIdenticalSources
identicalSources	IdenticalSources

83.2.5 SourceProperty.nonIdenticalSources

empty struct

83.2.6 SwitchingMode

InputRedundancy switching mode.

•

enum

REVERTING	allows stream source switching whenever the main is back. •
FLOATING	allows switching to another input and use it as long as it's error free.
OFF	Input Redundancy is turned off, no switching will takeplace.

83.2.7 TsSource

TS video source parameters.
 @param sources a map of the current stream source and their uuids. a MAIN stream source must always be present.
 @param inputRedundancy the input redundancy configuration parameters.
 @param subtitlePriorityList List of SubtitleComponentPattern describing which subtitle component is the most important.

struct

sources	map from streamSource.StreamSource to UUID
inputRedundancy	optional InputRedundancy
subtitlePriorityList	list of subtitling.SubtitleComponentPattern

84 vancProfile (1.7)

84.1 Overview

```
# Changelog:

## 1.7
- **Changed**
  - Moved `Dpi` and `DpiSource` to its own file.

## 1.6
- **Added**
  - New `horizontalAlignment` member to `DvbSubtitle`.
- **Changed**
  - `subtitling` from 1.1 to 1.2.

## 1.5
- **Added**
  - enum `DpiSource`
  - member `source` to Dpi

## 1.4
- **Added**
  - type `DvbSubtitle`
  - member `dvbSubtitle` to VancParams
  - value `DVBSUBTITLE` to VancParamsEnum
```

84.2 Command Reference

84.2.1 GetVancProfiles

- message **GetVancProfiles.Request**
- message **GetVancProfiles.Response**
- message **GetVancProfiles.Failure**

84.2.2 SetVancProfiles

- message **SetVancProfiles.Request**
- message **SetVancProfiles.Response**
- message **SetVancProfiles.Failure**

84.2.3 DeleteVancProfiles

- message **DeleteVancProfiles.Request**
- message **DeleteVancProfiles.Response**
- message **DeleteVancProfiles.Failure**

84.3 Type Reference

84.3.1 Anc

Anc.

struct

did	int vanc did.
sdid	int vanc sdid.
line	optional int line selection.

84.3.2 ClosedCaption

empty **struct**

84.3.3 DataSelection

Specifies which ancillary data packets are included in the SMPTE 2038 PID. An empty variant is treated as if auto was specified.

variant

manual	DataSelection.manual Only ancillary data that matches the provided list of DID/SDID pairs is used (up to 4 pairs supported).
auto	DataSelection.auto Use any incoming data.

84.3.4 DataSelection.auto

empty **struct**

84.3.5 DataSelection.manual

struct

input	list of Anc
-------	---------------------------

84.3.6 DeleteVancProfiles.Failure

empty **struct**

84.3.7 DeleteVancProfiles.Request

struct

ids	list of UUID
-----	----------------------------

84.3.8 DeleteVancProfiles.Response

empty **struct**

84.3.9 DvbSubtitle

DVB subtitle

struct

teletext	Teletext EBU teletext to be converted to DVB subtitles
horizontalAlignment	subtitling.HorizontalAlignment

84.3.10 EN301775

struct

teletext	optional Teletext
vps	optional Vps
wss	optional Wss
cc	optional ClosedCaption
dataId	optional int

84.3.11 GetVancProfiles.Failure

empty **struct**

84.3.12 GetVancProfiles.Request

empty **struct**

84.3.13 GetVancProfiles.Response

struct

data	map from UUID to VancProfile
info	GetVancProfiles.Response.info

84.3.14 GetVancProfiles.Response.info

struct

totalEntries	int
--------------	------------

84.3.15 SetVancProfiles.Failure

empty **struct**

84.3.16 SetVancProfiles.Request

struct

data	map from UUID to VancProfile
------	---

84.3.17 SetVancProfiles.Response

empty **struct**

84.3.18 Smpte2038

Smpte 2038.

struct

dataSelection	DataSelection modes of ANC data selection.
delay	int adds offset of x ms to PTS.
maxBitrate	optional int sets a max bitRate.

84.3.19 Teletext

struct

input	VancStandard
teletext	list of TeletextPsi
lineFilter	list of int

84.3.20 TeletextPsi

struct

page	int
language	string
teletextType	TeletextType

84.3.21 TeletextType

enum

START
SUBTITLE
HEARING_IMPAIRED

84.3.22 VancParams

Configuration of the type of ancillary data in transport stream

variant

dpi	dpi.Dpi Digital program insertion messages according to scte-35
en301775	EN301775 VBI in DVB according to ETSI EN 301 775

smpte2038

Smpte2038

Ancillary data packets in a transport stream according to SMPTE ST 2038

dvbSubtitle

DvbSubtitle

DVB subtitles according to ETSI EN 300 743

84.3.23 VancParamsEnum**enum**

DPI

EN301775

SMPTE2038

DVBSUBTITLE

84.3.24 VancProfile**VancProfileStruct****84.3.25 VancProfileStruct****struct**

label

string

vancParams

VancParams**84.3.26 VancStandard****enum**

OP_47

SMPTE_2031

VBI

84.3.27 Vpsempty **struct****84.3.28 Wss**empty **struct**

85 vancStatus (1.13)

85.1 Overview

Changelog:

1.13

- **Changed**
 - `vancProfile` from 1.6 to 1.7
 - Moved `DpiCommand`, `DpiInput`, and `DpiStatus` to its own file.

1.12

- **Changed**
 - vancProfile from 1.5 to 1.6

1.11

- **Changed**
 - sourceIdentifier member in dpiInput.

1.10

- **Changed**
 - dpiIn member in VancStatus from DpiInput to list(DpiInput).
 - Added timeSinceLastReceived to DpiInput.
 - Added lastGeneratedCommand to DpiInput.
 - Added DpiCommand type

1.9

- **Changed**
 - added value timeSinceLastReceived to DpiStatus
 - added value timeSinceLastGenerated to DpiStatus
 - added value lastCommandType to DpiStatus

1.8

- **Changed**
 - vancProfile from 1.4 to 1.5

1.7

- **Changed**
 - added value dvbSubtitle to VancComponentStatus
 - added value DVBSUBTITLESTATUS to VancComponentStatusEnum
 - changed function VancComponentStatus2Enum to accomodate DVBSUBTITLESTATUS
- **Added**
 - struct DvbSubtitleStatus

85.2 Type Reference

85.2.1 AspectRatioInput

struct

standard	aspectRatio.AspectRatioStandard
format	afd.AfdCode
is16x9	bool

85.2.2 DataIdentifier

The Data Identifier (DID) and Secondary Data Identifier (SDID) words which together indicate a type of ancillary data.

struct

did	int The Data Identifier word.
sdid	int The Secondary Data Identifier word.
lineNum	int The video line number of the DID/SDID.

85.2.3 DvbSubtitleStatus

DVB subtitle status

struct

enabled	bool Set to false if the feature is prevented from running (by licences, invalid config etc)
timeSinceLastReceived	int Seconds since last valid EBU subtitle input
timeSinceLastGenerated	int Seconds since last generated DVB subtitle

85.2.4 En301775Status

struct

txtStatus	optional TeletextStatus
-----------	--------------------------------

85.2.5 Smp2038Status

Smp2038 Status.

struct

enabled	bool
running	bool

dataIdentifiers **list** of **DataIdentifier**

85.2.6 TeletextInput

struct

input	vancProfile.VancStandard
linesIn	list of int
invalidPackets	int
received	bool

85.2.7 TeletextStatus

struct

standard	vancProfile.VancStandard
enabled	bool
running	bool
linesOut	list of int

85.2.8 VancComponentStatus

variant

dpi	dpiStatus.DpiStatus
en301775	En301775Status
smpte2038	Smpte2038Status
dvbSubtitle	DvbSubtitleStatus

85.2.9 VancComponentStatusEnum

enum

DPISTATUS	
En301775STATUS	
SMPTE2038STATUS	
DVBSUBTITLESTATUS	

85.2.10 VancStatus

struct

ttxtIn	list of TeletextInput
dpiIn	list of dpiStatus.DpiInput
arIn	list of AspectRatioInput
outputs	map from int to VancComponentStatus
sources	map from int to set of int
dataIdentifiers	list of DataIdentifier

86 videoProfile (2.11)

86.1 Overview

```
# Changelog:
## 2.11
- **Changed**
  - `rawVideo` from 1.4 to 1.5

## 2.10
- **Added**
  - extendedSar bool to videoProfileStruct
```

86.2 Command Reference

86.2.1 GetVideoProfiles

- message **GetVideoProfiles.Request**
- message **GetVideoProfiles.Response**
- message **GetVideoProfiles.Failure**

86.2.2 SetVideoProfiles

- message **SetVideoProfiles.Request**
- message **SetVideoProfiles.Response**
- message **SetVideoProfiles.Failure**

86.2.3 DeleteVideoProfiles

- message **DeleteVideoProfiles.Request**
- message **DeleteVideoProfiles.Response**
- message **DeleteVideoProfiles.Failure**

86.3 Type Reference

86.3.1 Afd

enum

Off
Passthrough

86.3.2 AspectRatio

enum

AR_TRANSPARENT
AR_16x9
AR_14x9
AR_4x3
AR_AFD
AR_WSS

```

AR_WSS18
AR_VideoIndex_11_324
AR_VideoIndex_8_321

```

86.3.3 AvcLevel

enum

```

AUTO
LEVEL_3
LEVEL_3p1
LEVEL_3p2
LEVEL_4
LEVEL_4p1
LEVEL_4p2
LEVEL_5
LEVEL_5p1

```

86.3.4 AvcParameters

struct

profile	AvcProfile
level	AvcLevel
codingMode	CodingMode
idrFrequency	int
cabac	bool

86.3.5 AvcProfile

enum

```

MAIN
HIGH
HIGH10
HIGH42210
CONSTRAINED_BASELINE

```

86.3.6 Bitrate

variant

cbr	int
cvbr	Cvbr

86.3.7 CodecParameters

variant

hevc	HevcParameters
avc	AvcParameters
mpeg2	Mpeg2Parameters

86.3.8 CodingMode

enum

AUTO
FRAME
FIELD
PAFF
MBAFF

86.3.9 Cvbr

struct

maxBitrate	int
averageBitrate	int

86.3.10 DeleteVideoProfiles.Failure

empty struct

86.3.11 DeleteVideoProfiles.Request

struct

ids	list of UUID
-----	--------------

86.3.12 DeleteVideoProfiles.Response

empty struct

86.3.13 GetVideoProfiles.Failure

empty struct

86.3.14 GetVideoProfiles.Request

empty struct

86.3.15 GetVideoProfiles.Response

struct

data	map from UUID to VideoProfile
info	GetVideoProfiles.Response.info

86.3.16 GetVideoProfiles.Response.info

struct

totalEntries	int
--------------	-----

86.3.17 GopMode

enum

Static	
Dynamic	

86.3.18 GopParameters

struct

size	int
gopMode	GopMode
structure	GopStructure
maxBframes	int
ldb	bool
hierarchical	bool

86.3.19 GopStructure

enum

I_ONLY	
IP	
IPB	
GDR	

86.3.20 HevcLevel

enum

AUTO	
LEVEL_3	
LEVEL_3p1	
LEVEL_4	
LEVEL_4p1	
LEVEL_5	
LEVEL_5p1	

86.3.21 HevcParameters

struct

profile	HevcProfile
---------	-------------

level	HevcLevel
tier	HevcTier
idrFrequency	int

86.3.22 HevcProfile

enum

MAIN
MAIN10
MAIN42210

86.3.23 HevcTier

enum

Main
High

86.3.24 Latency

enum

NORMAL
LOW
ULL

86.3.25 Mpeg2Level

enum

MAIN
HIGH

86.3.26 Mpeg2Parameters

struct

profile	Mpeg2Profile
level	Mpeg2Level
codingMode	CodingMode
openGop	bool

86.3.27 Mpeg2Profile

enum

MAIN
PROFILE_422P

86.3.28 ProfileType

enum

ENCODER	
DECODER	
TRANSCODER	

86.3.29 ScanningMode

enum

INTERLACED	
PROGRESSIVE	

86.3.30 SetVideoProfiles.Failure

empty **struct**

86.3.31 SetVideoProfiles.Request

struct

data	map from UUID to VideoProfile
------	--

86.3.32 SetVideoProfiles.Response

empty **struct**

86.3.33 VideoCodec

enum

HEVC	
AVC	
MPEG2	
JPEG2000	

86.3.34 VideoProfile

VideoProfileStruct

86.3.35 VideoProfileStruct

struct

profileType	ProfileType
label	string
bitDepth	int
latency	Latency

resolution	VideoResolution
codec	VideoCodec
bitrate	Bitrate
cparams	CodecParameters
ar	AspectRatio
fallbackAr	AspectRatio
gop	GopParameters
afd	Afd
wssBlanking	bool
extendedSar	bool

86.3.36 VideoResolution

struct

horizontal	rawVideo.HorizontalResolution
vertical	rawVideo.VerticalResolution
chromaSampling	rawVideo.ChromaSampling
scan	ScanningMode
fps	rawVideo.FrameRate
followInput	bool

87 videoStatus (2.15)

87.1 Overview

Video status types and objects. This IDCL is part of the IDCL sent to the GUI

Changelog:

2.15

- **Changed**
 - 'videoProfile from 2.10 to 2.11

2.14

- **Changed**
 - 'videoProfile from 2.9 to 2.10
 - VuiColorimetry added to Es

2.13

- **Changed**
 - colorProfile to videoUsabilityInfo

2.12

- **Changed**
 - videoProfile from 2.8 to 2.9
 - ccPresent added to Es

2.11

- **Changed**
 - Ts.bitRate changed from int to bigint to allow for high bitrates, which will be necessary for UHD

87.2 Type Reference

87.2.1 Es

Status for coded video.

struct

valid	bool
hSize	int
vSize	int
bitRate	int
cpb	optional int
bitDepth	int
chroma	int
codec	videoProfile.VideoCodec
aspectRatio	optional videoProfile.AspectRatio
framerate	optional rawVideo.FrameRate
scan	optional videoProfile.ScanningMode

colorDescPresent	bool
colorPrimaries	videoUsabilityInfo.ColorSpace
transferChar	videoUsabilityInfo.TransferFunc
matrixCoeffs	videoUsabilityInfo.ColorSpace
idrOnScte35	optional int
ccPresent	bool
vuiColorimetry	optional VuiColorimetry

87.2.2 InputStatus

struct

id	InstanceId
input	VideoFormat

87.2.3 InstanceId

int

87.2.4 OutputStatus

struct

id	InstanceId
output	VideoFormat

87.2.5 Raw

struct

valid	bool
numFrames	int
videoMode	optional VideoModes
framerate	optional rawVideo.FrameRate
chroma	optional int
bitDepth	optional int
vitcPresent	bool
vitcLineNr	optional int
ccPresent	bool
timecode	optional timecodeStatus.Timecode

87.2.6 Ts

struct

bitRate	bigint
ccError	int
pcrJitterError	optional bool
pcrDisc	optional bool

pcrIntervalErr	optional bool
es	Es
pid	optional int

87.2.7 VideoFormat

variant

raw	Raw
es	Es
ts	Ts

87.2.8 VideoModes

enum

V_UNKNOWN
V_480i
V_576i
V_720p
V_1080i
V_1080p
V_2160p

87.2.9 VideoStatus

struct

input	InputStatus
output	OutputStatus

87.2.10 VuiColorimetry

struct

colorPrimaries	int
transferChar	int
matrixCoeffs	int

88 videoUsabilityInfo (1.1)

88.1 Overview

```
# Changelog
## 1.1
- **Changed**
  Added BT_601 to `ColorSpace`
  Added BT_601 to `TransferFunc`
```

88.2 Type Reference

88.2.1 ColorSpace

ColorSpace

ColorSpace is a set of colour coordinates for what each of red, green, blue and white “mean”

enum

UNDEFINED	
BT_709	
BT_601	
BT_2020	

88.2.2 TransferFunc

TransferFunc

Transfer function to convert video signals to light

enum

UNDEFINED	
BT_709	
BT_601	
BT_2010_SDR	
BT_2011_PQ	
BT_2011_HLG	

88.2.3 VuiParameters

VuiParameters

Video Usability Info

struct

colorSpace	ColorSpace A set of colour coordinates for what each of red, green, blue and white “mean”.
transfer	TransferFunc Transfer function

ColorSpace

A set of colour coordinates for what each of red, green, blue and white “mean”.

TransferFunc

Transfer function

ColorSpace

matrix

Specifies the matrix coefficients used to derive the luma and chroma from the RGB primaries.

89 vpidInfo (1.0)

89.1 Type Reference

89.1.1 ColorInfo

struct

colorimetry	Colorimetry
transfer	TransferCharacteristics

89.1.2 Colorimetry

enum

REC_709	
COLOR_VANC	
UHD_TV	
UNKNOWN	

89.1.3 TransferCharacteristics

enum

SDR	
HLG	
PQ	
UNSPECIFIED	